- 概率与数理统计
  - 所有的学科都离不开概率与统计
- 曹圣彬
  - cccvolcano@m.scnu.edu.cn
- 教科书
  - Jay L. Devore, “Probability and statistics for engineering and the sciences (the 7th Edition)”
- 课外参考材料
  - Miller and Freund, “Probability and Statistics for Engineers (the 7th Edition)”
  - 盛骤, 谢式千, 潘承毅 “概率论与数理统计 (第4版)”, 高等教育出版社, 2008
- 考核方式
  - 课堂表现（出勤，作业等） 30%
  - 期末考试 （闭卷形式） 70%

Probability and Statistics

Overview

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Introduction

- Why do we need to study statistics and probability?

Introduction

- Why do we need to study statistics and probability?
  - Dice rolling game



Introduction

- Why do we need to study statistics and probability?
  - Dice rolling game
  - What if there are only single number on all six sides of the dice?





Introduction

- Why do we need to study statistics and probability?
  - Dice rolling game
  - What if there are only single number on all six sides of the dice?





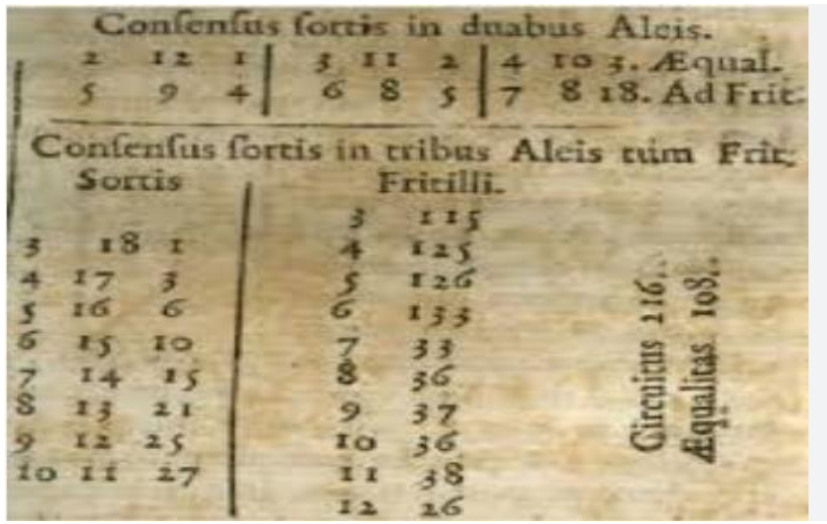
Introduction

- Uncertainty
  - Statistics and probability stem from uncertainty
  - Both fields are fundamentally concerned with quantifying, modeling, and making decisions based on uncertainty

Introduction

- Uncertainty
  - Statistics and probability stem from uncertainty
  - Both fields are fundamentally concerned with quantifying, modeling, and making decisions based on uncertainty
    - Statistics and probability came from gambling in 16th centrury Europe
    - The first book on statistics and probability is “Book on Games of Chance” written by Liber de Ludo Aleae in 1564 to analyze the gambling.





Book on Games of Chance

Introduction

- Uncertainty
  - Statistics and probability stem from uncertainty
  - Both fields are fundamentally concerned with quantifying, modeling, and making decisions based on uncertainty
  - In english language, can be expressed as “likely”, “maybe”
  - Any better way to describe?

Introduction

- Probability
  - A formal and mathematical way to describe uncertainty
    - The result of dice rolling is likely to be 5 - language
    - The probability that 5 side will appear is 1/6 - probability
  - Tossing a coin, what is the probability that head side appears?

Introduction

- Probability
  - A formal and mathematical way to describe uncertainty
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    - 50%

Introduction

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- Statistics
  - Related to sample
  - A method to draw some information from sample data set
  - The number of male students in this class is 35



Sample

  - How many males students are there in the entire university?

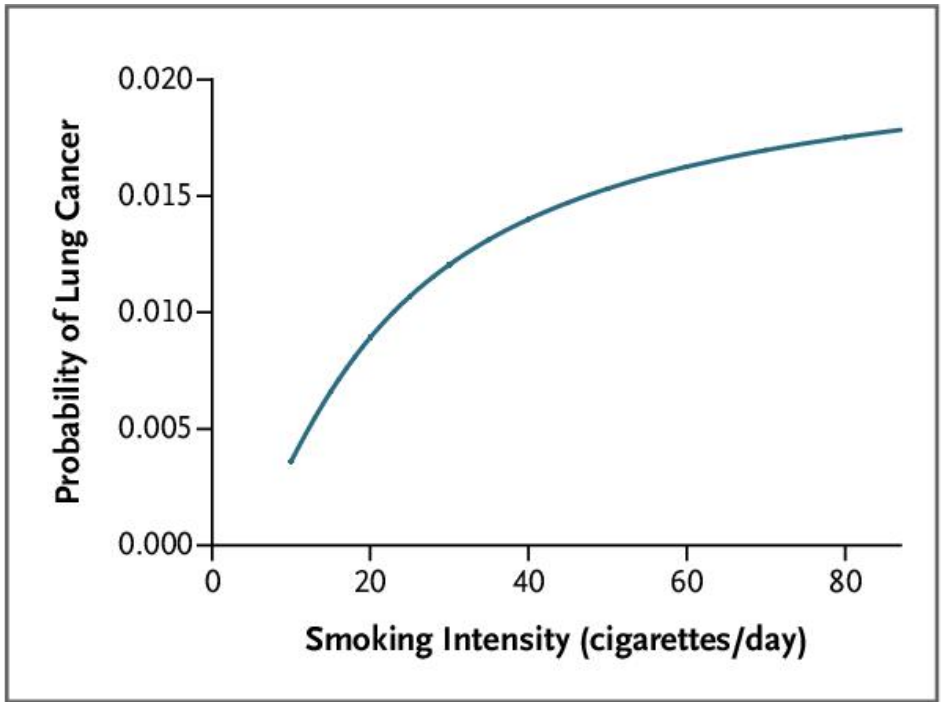
Introduction

- Weather forecast
  - Probability of raining

	M	T	W	TH	F	S	S
Chance of rainfall	70%	80%	90%	80%	60%	20%	0%
							

Introduction

- Medical treatment
  - The relationship between lung cancer probability and smoking





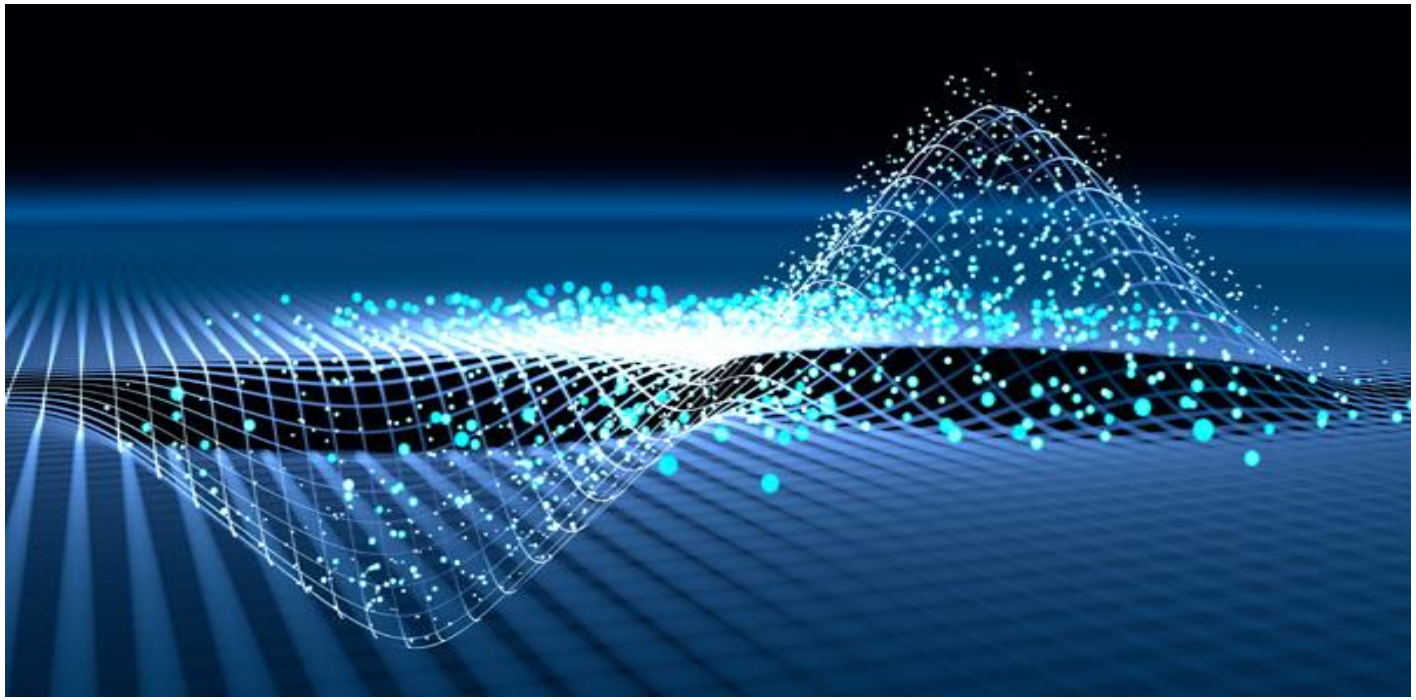
Introduction

- Market analysis
  - Economists analyze market trends and economic indicators to indicate the probability of investment success



Introduction

- Communication and electronics
  - Statistical methods are used to filter out noise from signals, improve data transmission quality



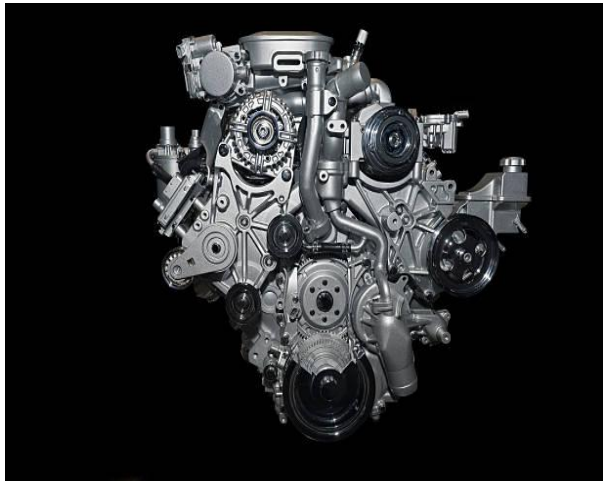
Introduction

- Engineering
  - Predict failure rate of a machine, helping in the design of more reliable products



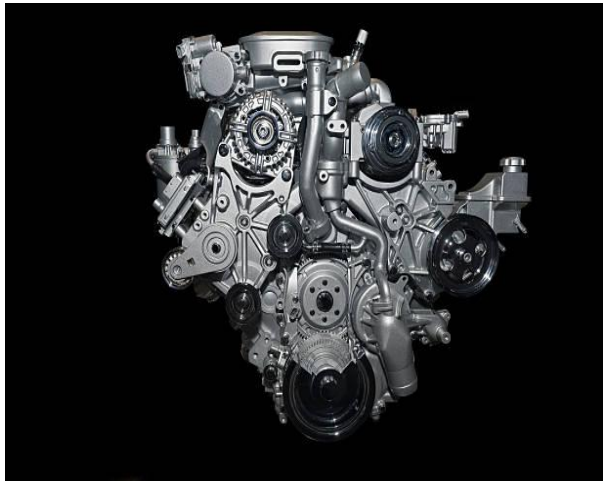
Introduction

- Markov chain analysis



Introduction

- Markov chain analysis
  - States
    - Fully operational (S1), degraded operational (S2), failuer (S3)



Introduction

- Markov chain analysis
  - States
    - Fully operational (S1), degraded operational (S2), failuer (S3)
  - Transition probability (statistics)

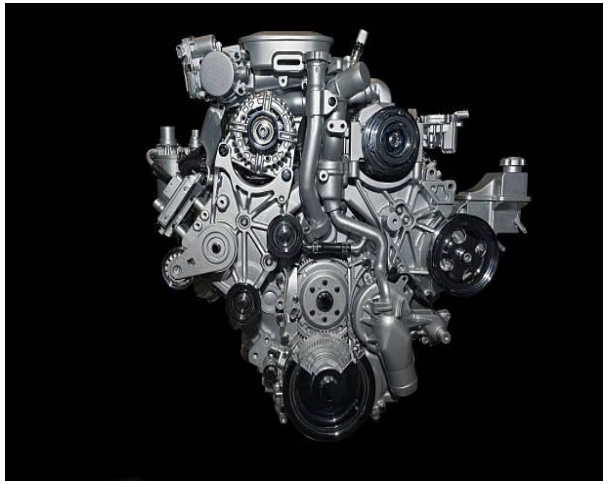
From S1 to S2 (degradation): Probability $P_{12} = 0.1$

From S1 to S3 (direct failure): Probability $P_{13} = 0.05$

From S2 to S1 (repair and return to full operation): Probability $P_{21} = 0.2$

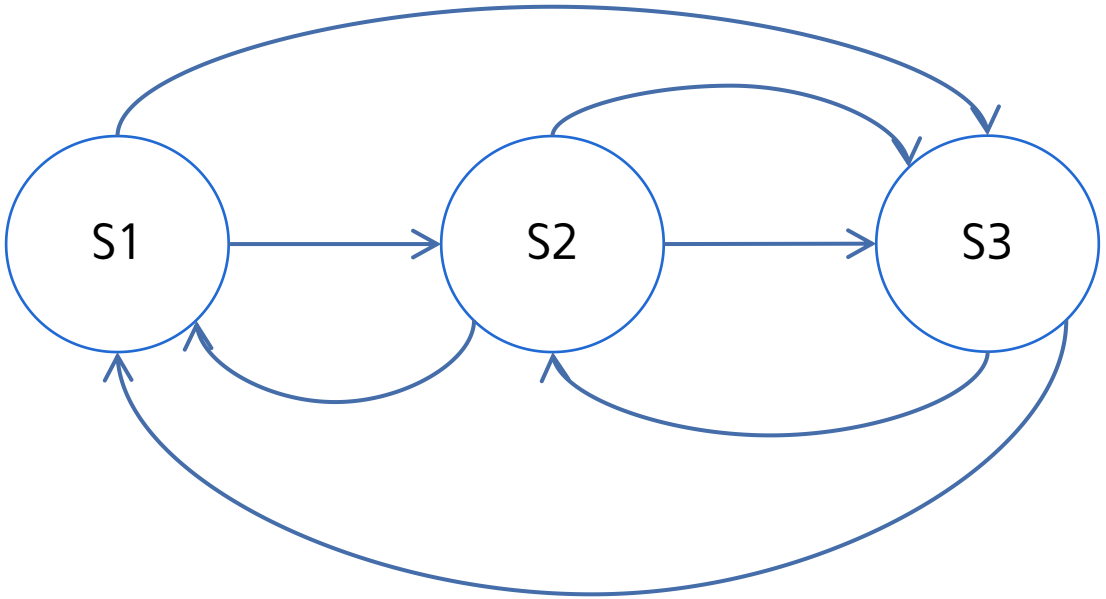
From S2 to S3 (failure from degraded state): Probability $P_{23} = 0.1$

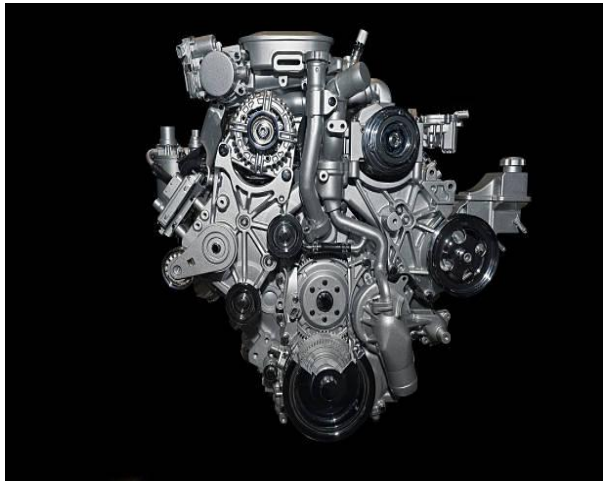
From S3 to S1 (repair from failure to full operation): Probability $P_{31} = 0.15$



Introduction

- Markov chain analysis
  - States
    - Fully operational (S1), degraded operational (S2), failuer (S3)
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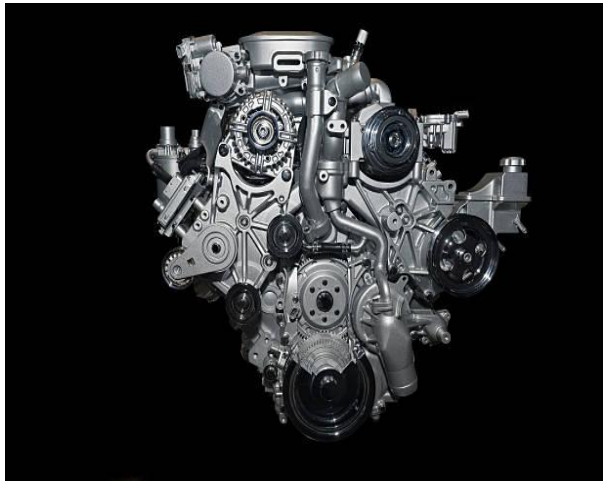
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  - Steady state analysis (probability)
    - The probability for the machine to be a specific state after long time

Introduction

- Markov chain analysis
  - States
    - Fully operational (S1), degraded operational (S2), failuer (S3)
  - Transition probability (statistics)

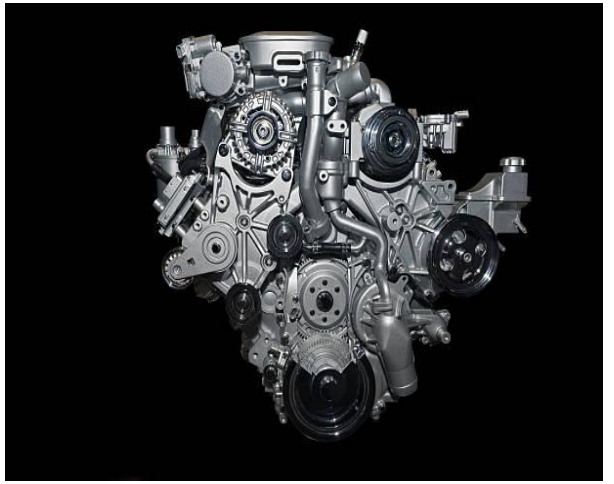
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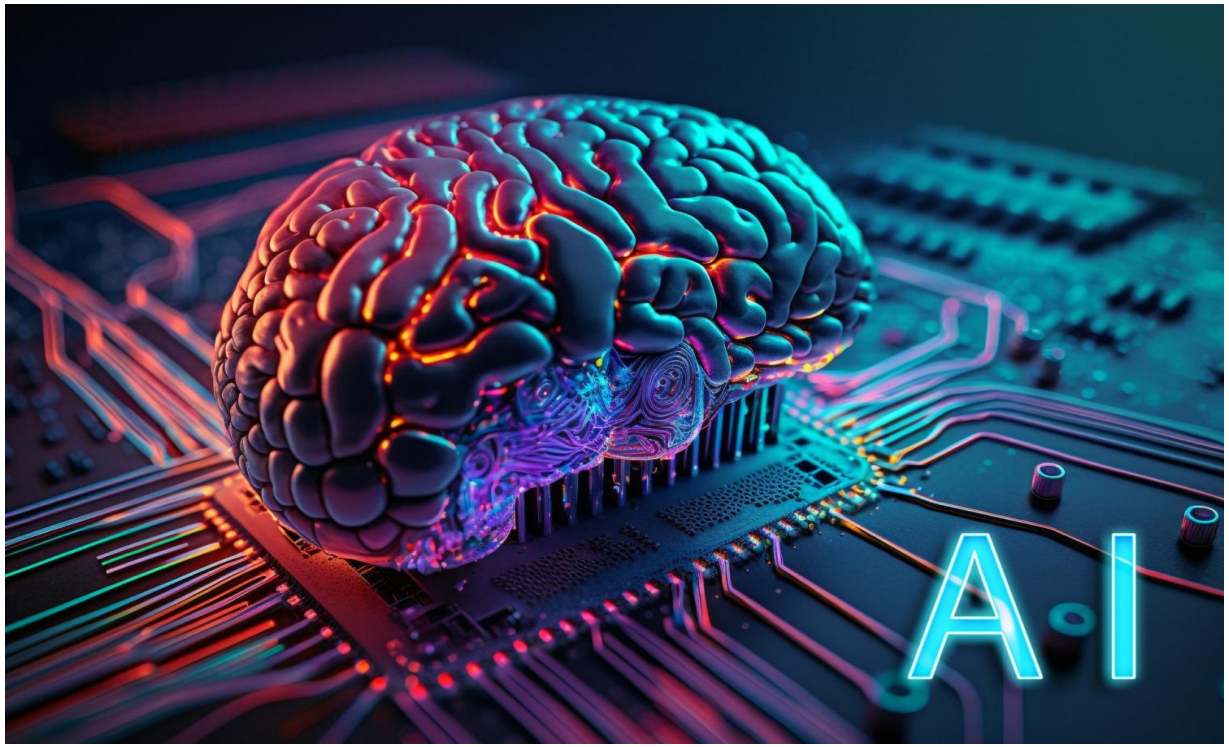


  - Steady state analysis (probability)
    - The probability for the machine to be a specific state after long time
  - Reliability
    - The reliability is the steady state probability of S1 + S2

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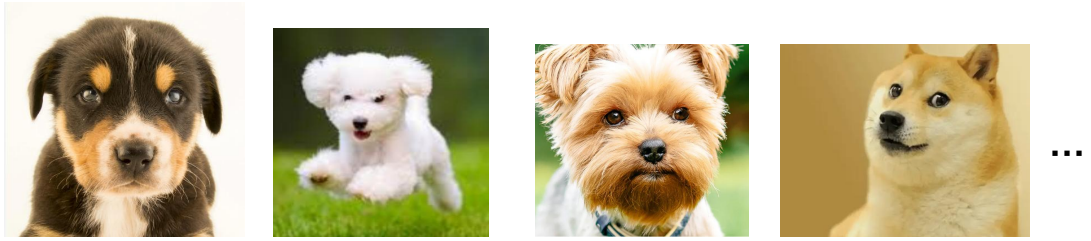
Introduction

- Artificial intelligence
  - AI relies heavily on statistical methods to learn from data and make predictions

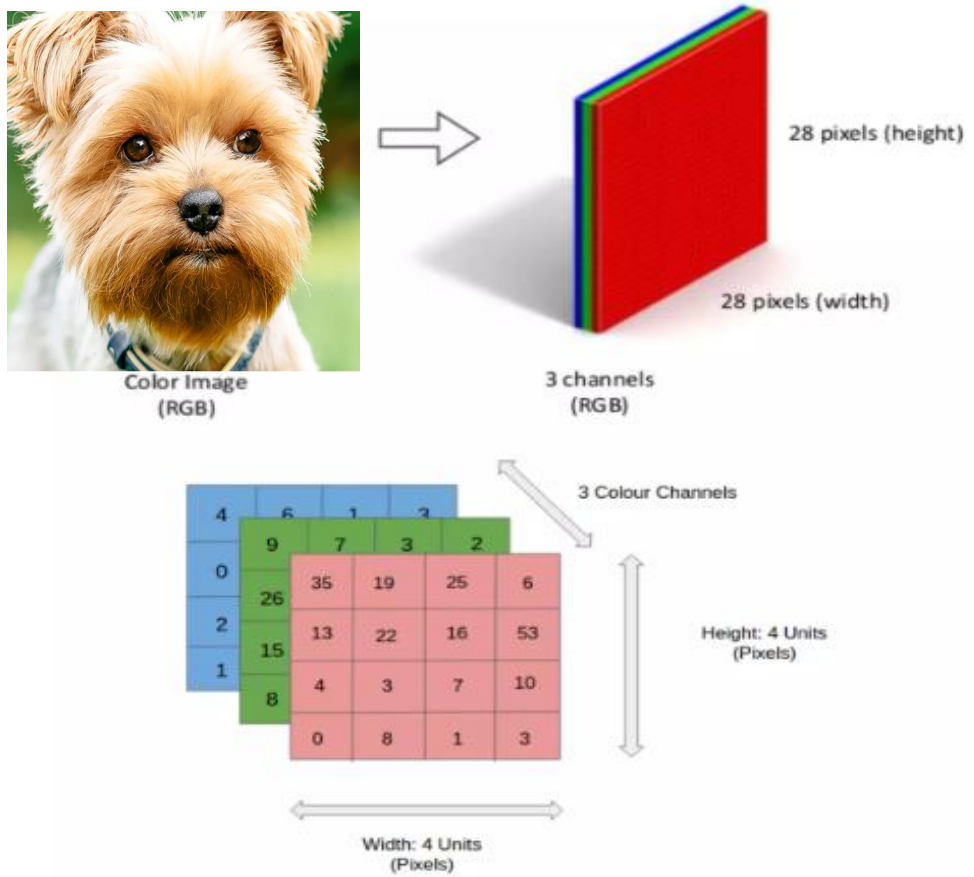


Introduction

- Deep learning
  - Image recognition

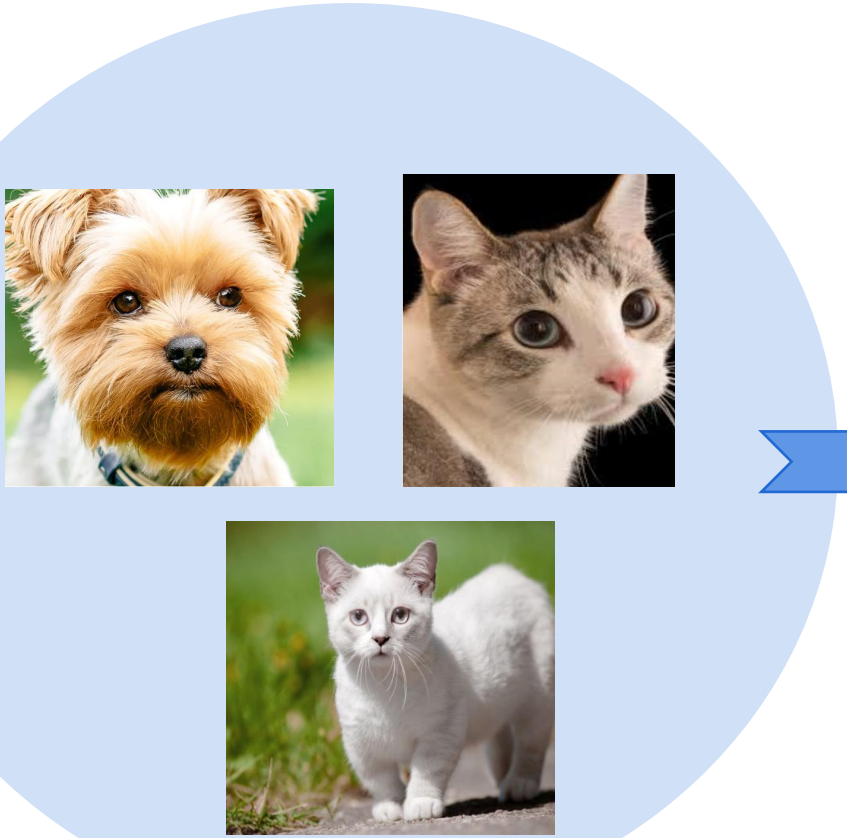




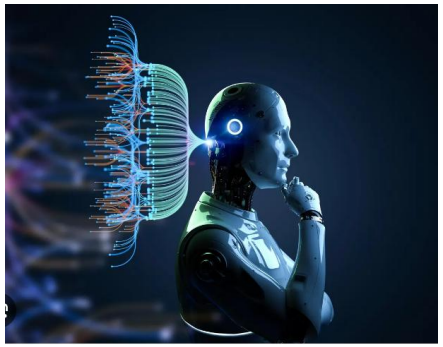


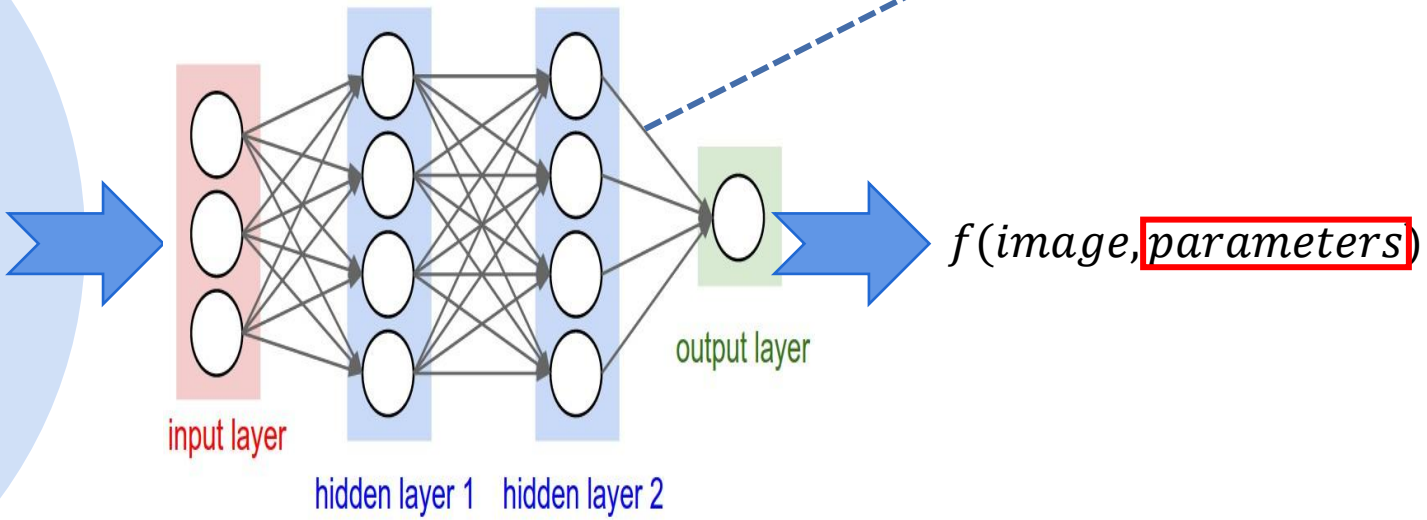
Introduction

- Deep learning
  - Training process



Samples



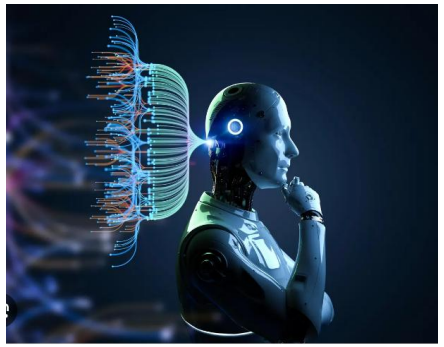


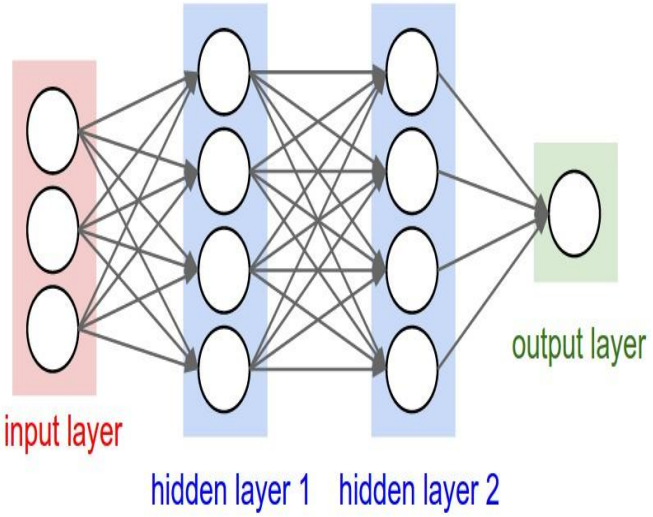
Deep neural network

Introduction

- Deep learning
  - Identification





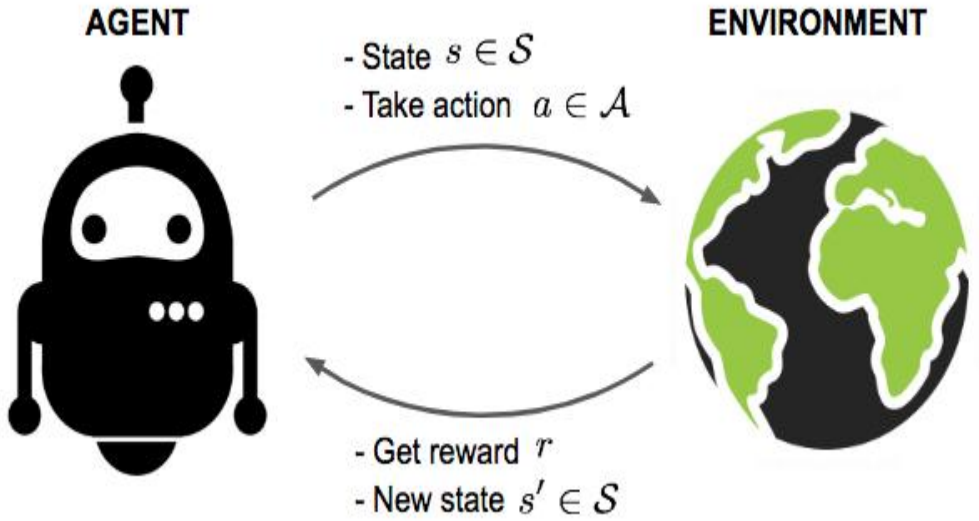


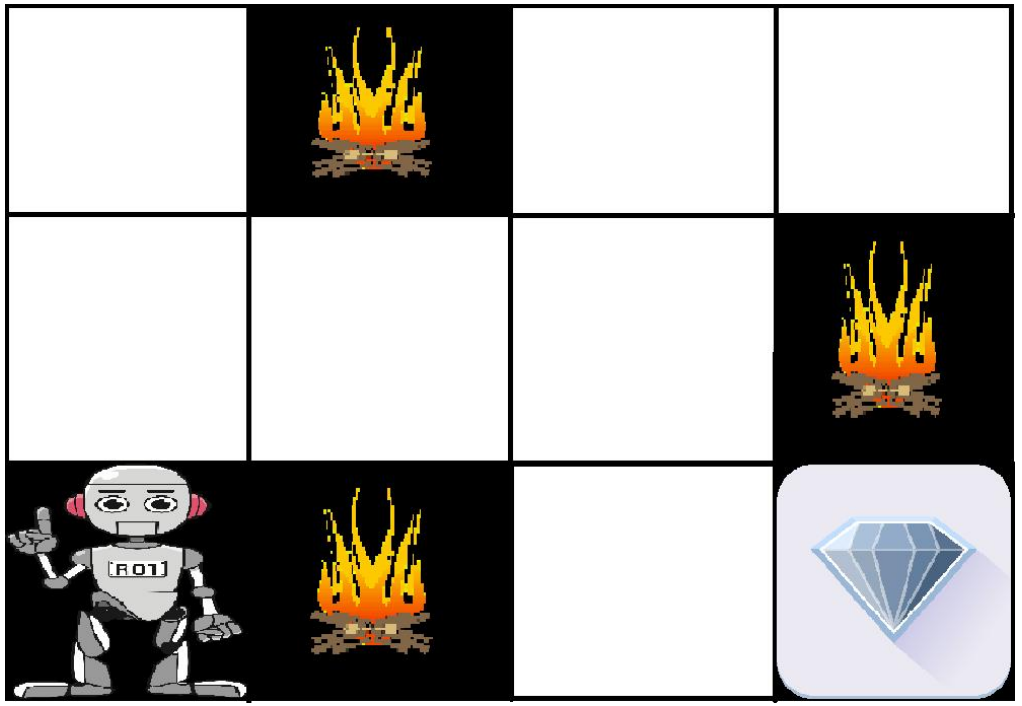
Probability for
the image to be dog

If probability > 0.5
  a dog
else
  a cat

Introduction

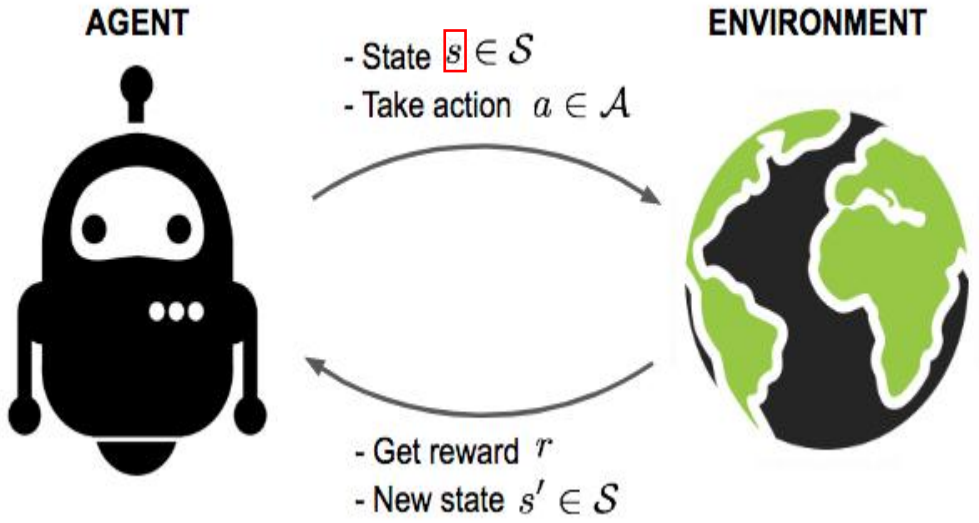
- Reinforcement learning

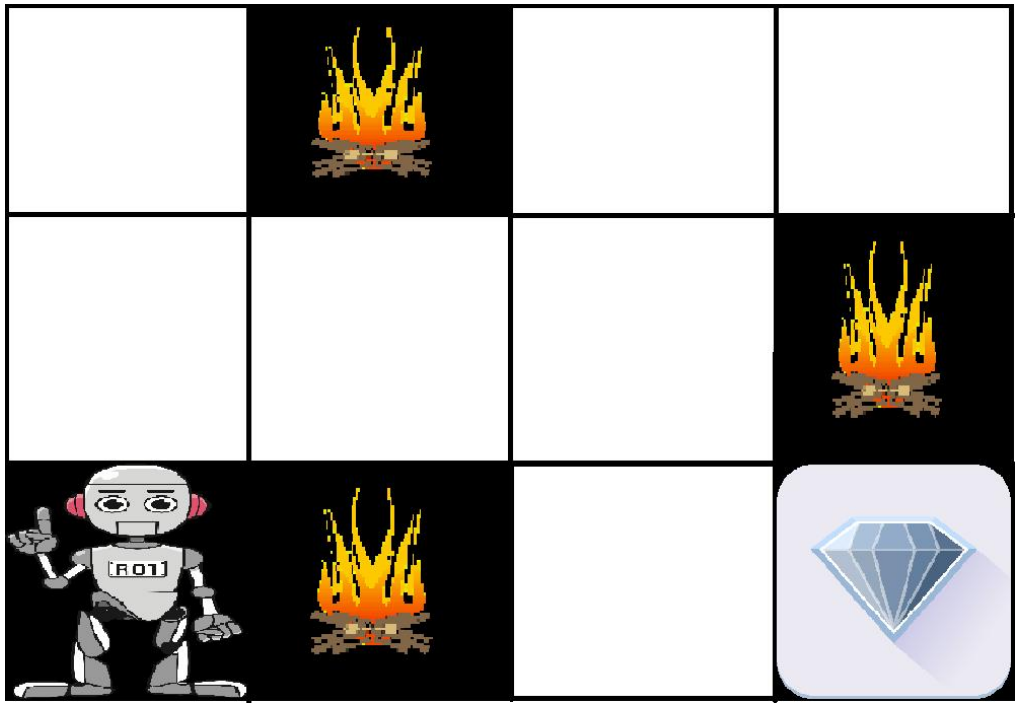




Introduction

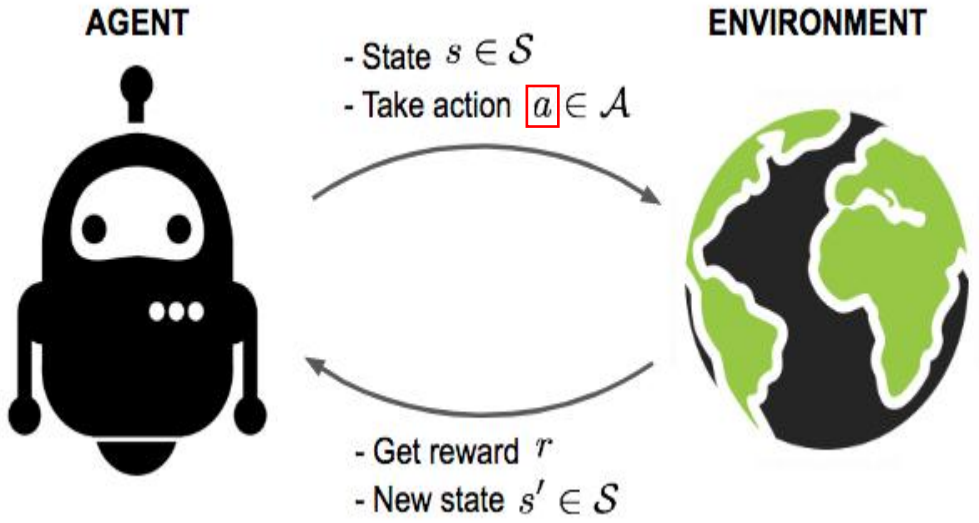
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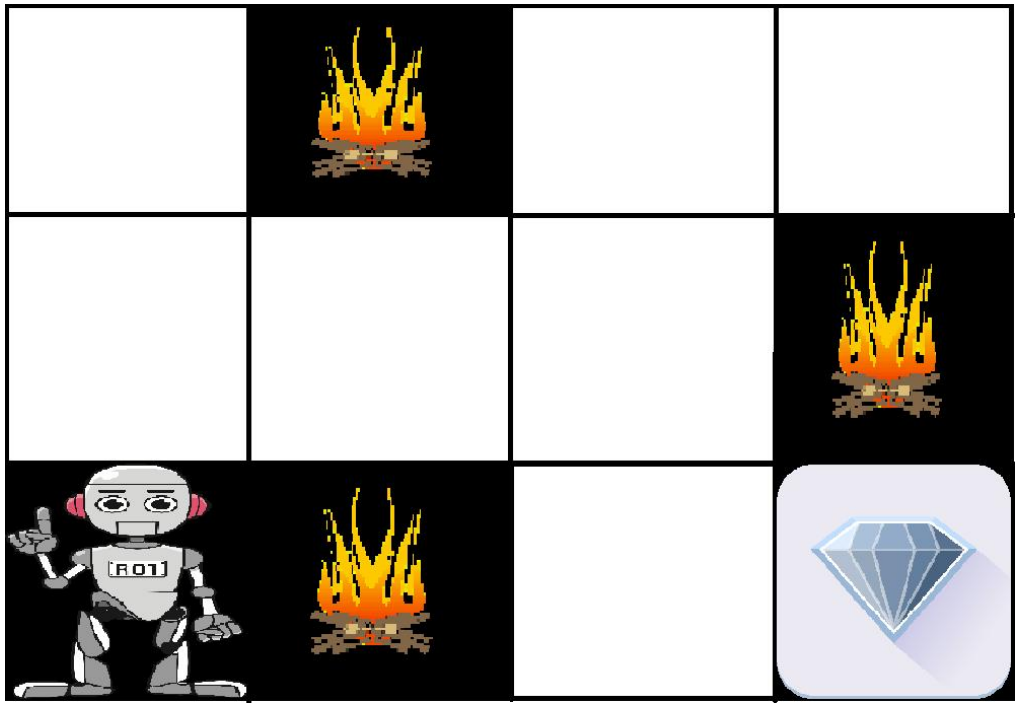




Introduction

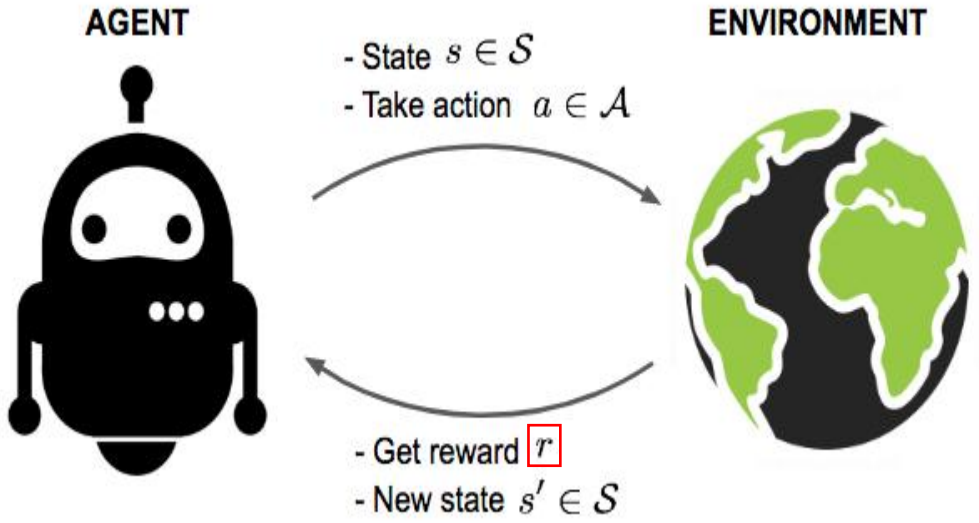
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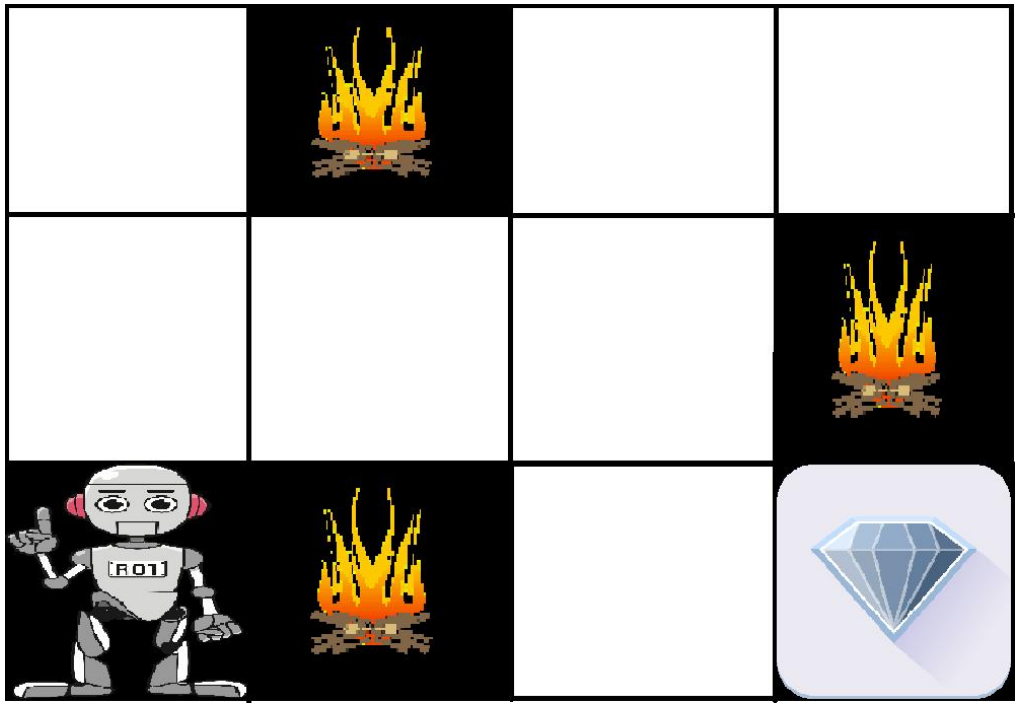




Introduction

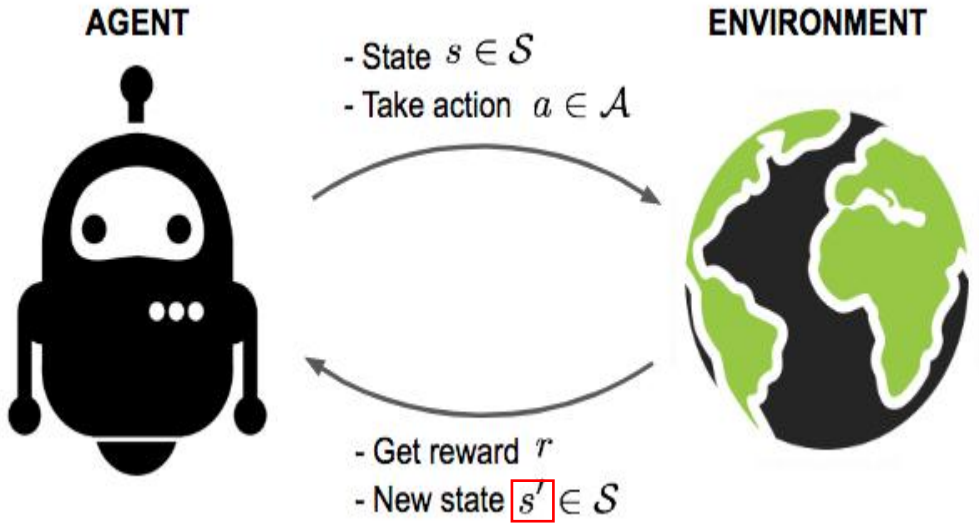
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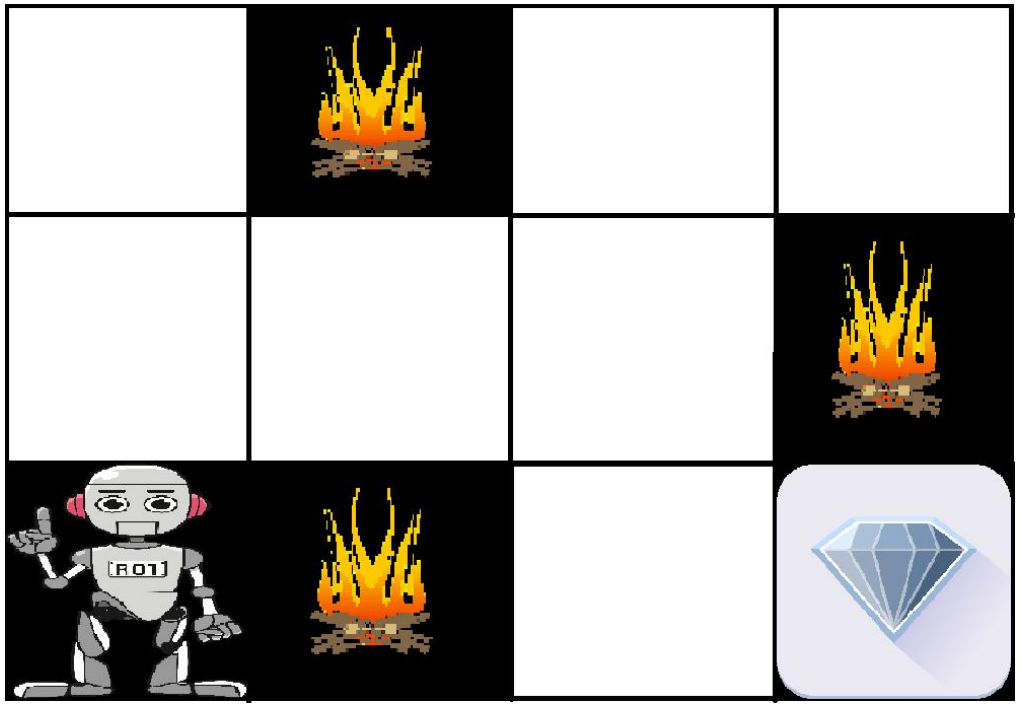




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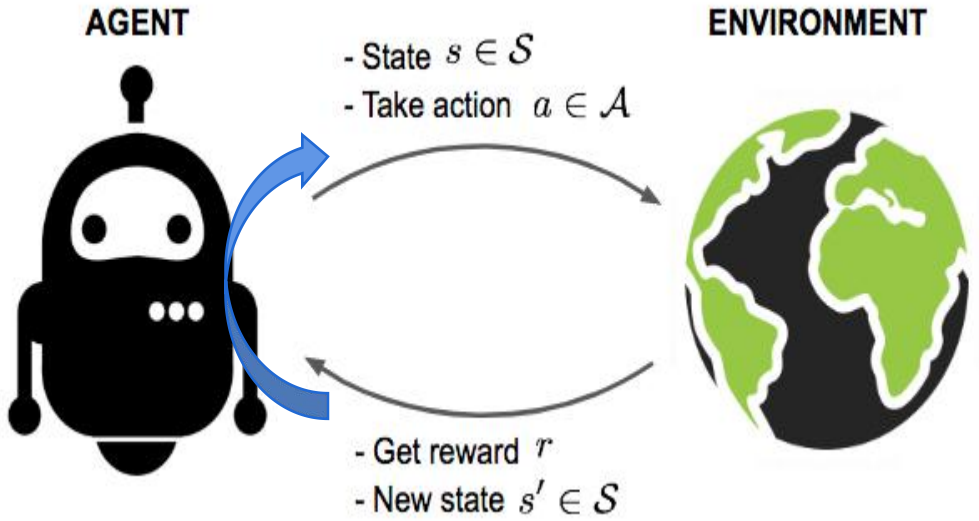
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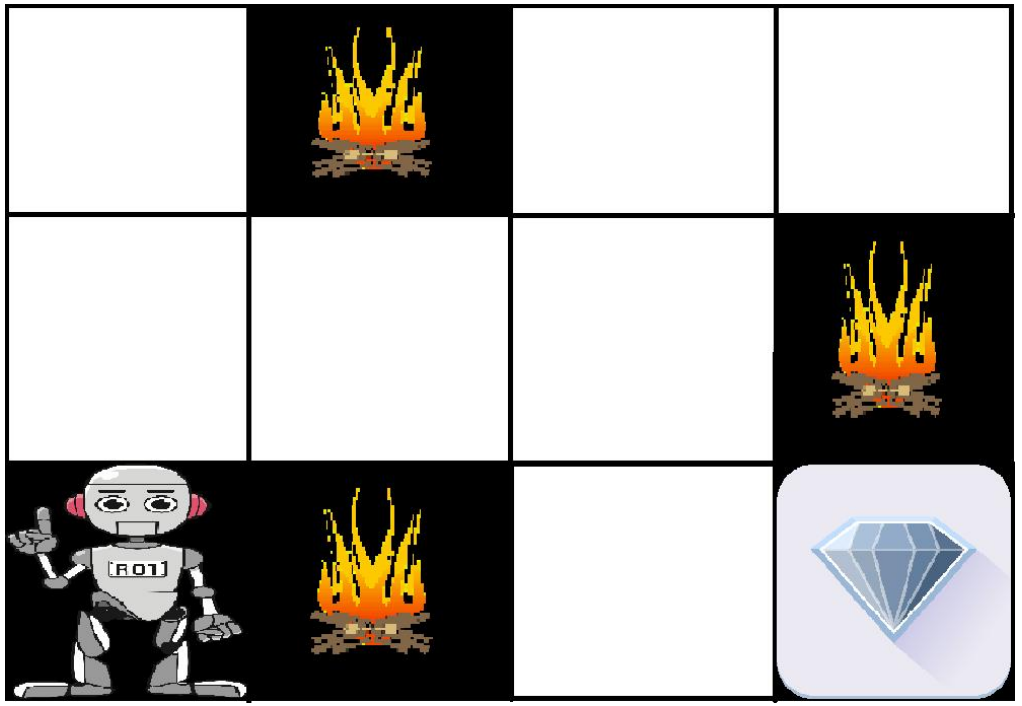




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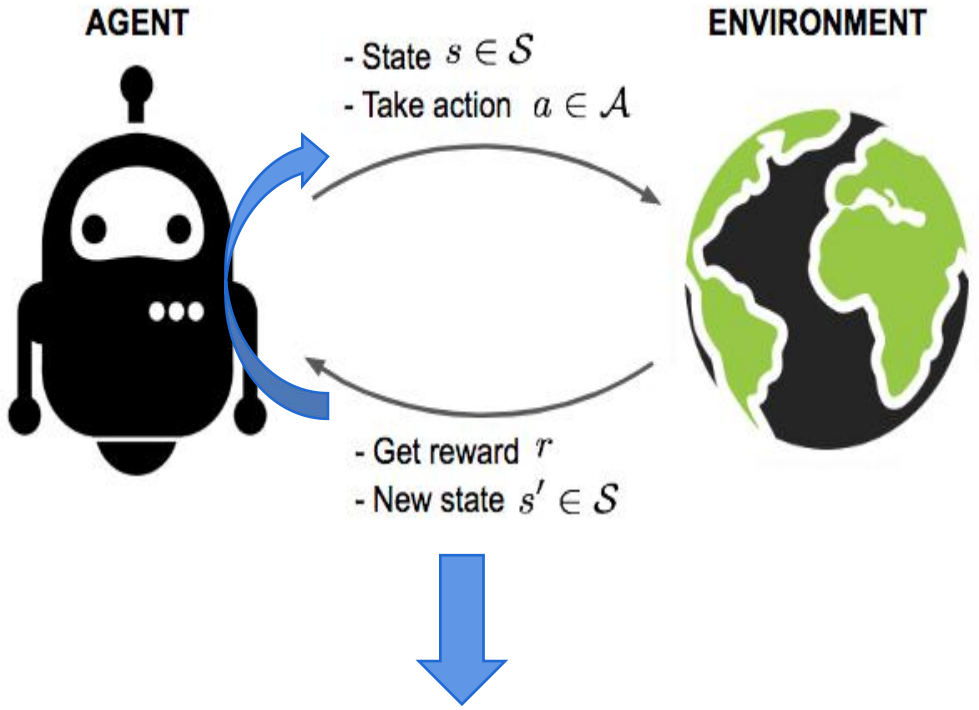
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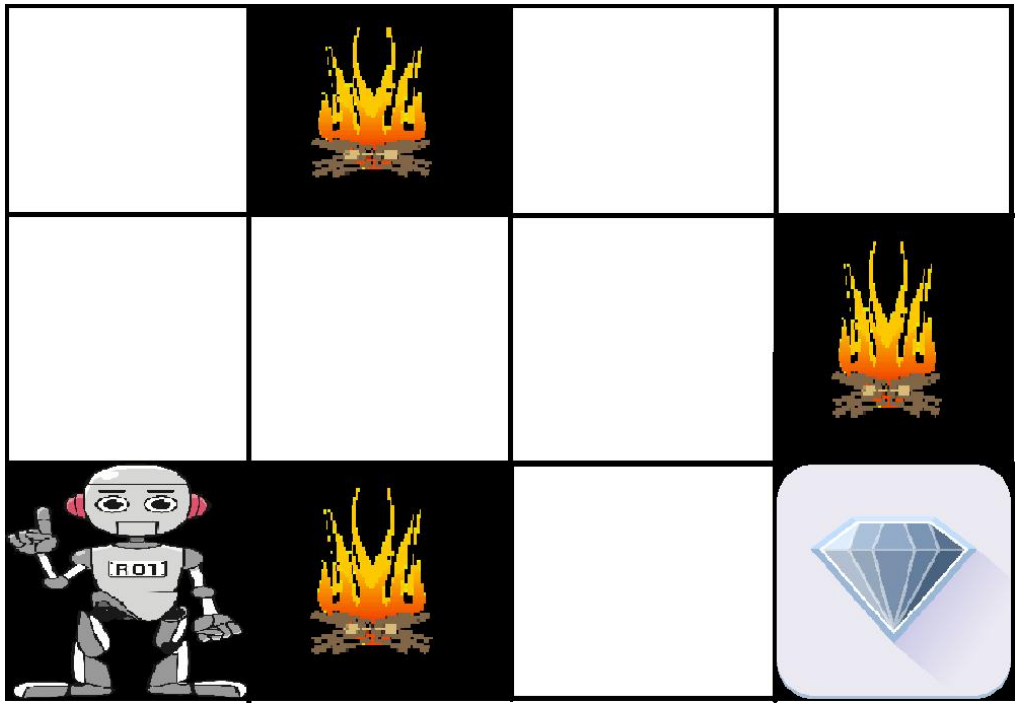




Introduction

- Reinforcement learning

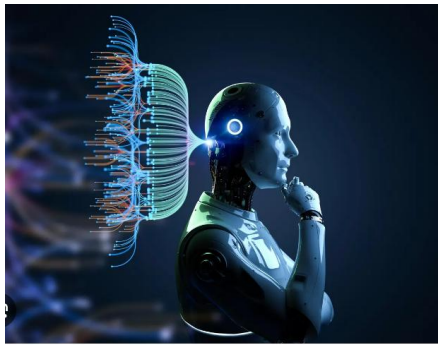




Training: Indicate the best action and give it the highest probability

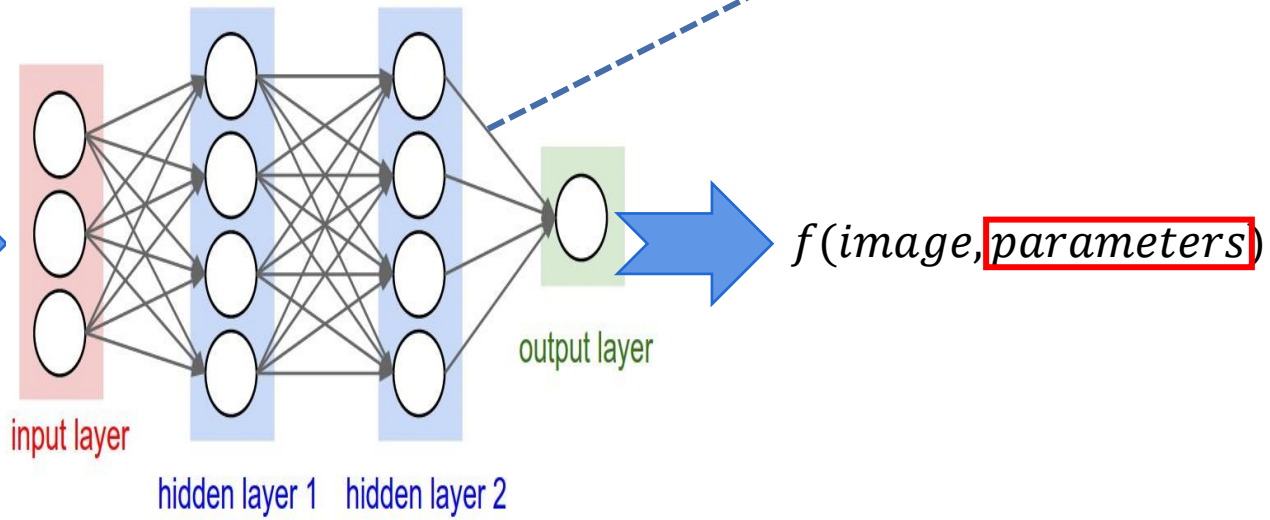
Introduction

- Deep learning
  - From different samples to learn best parameters





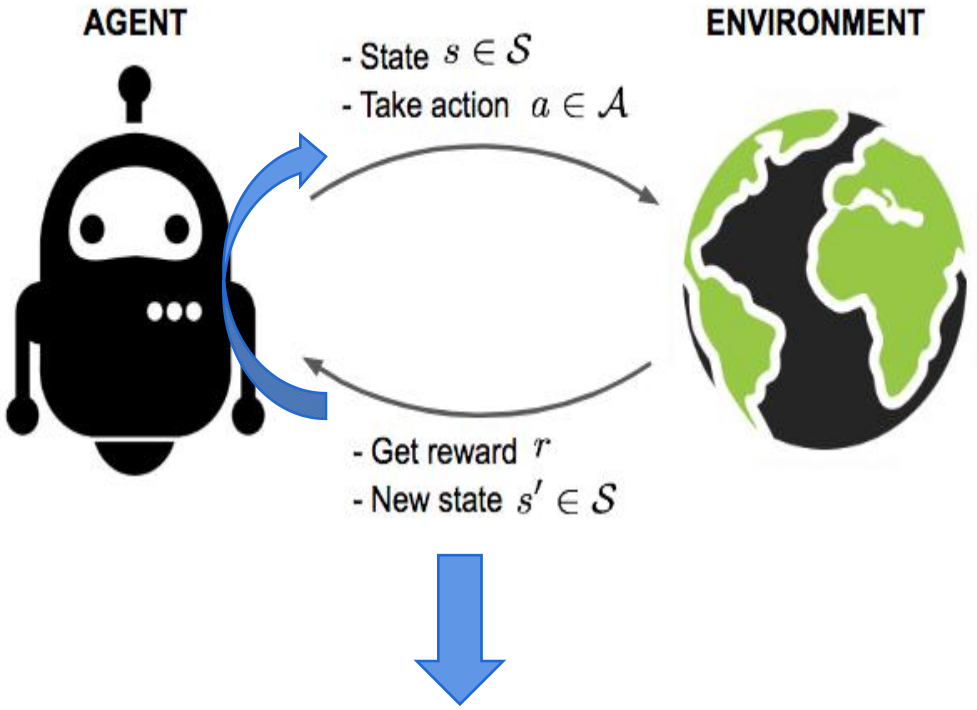
Samples

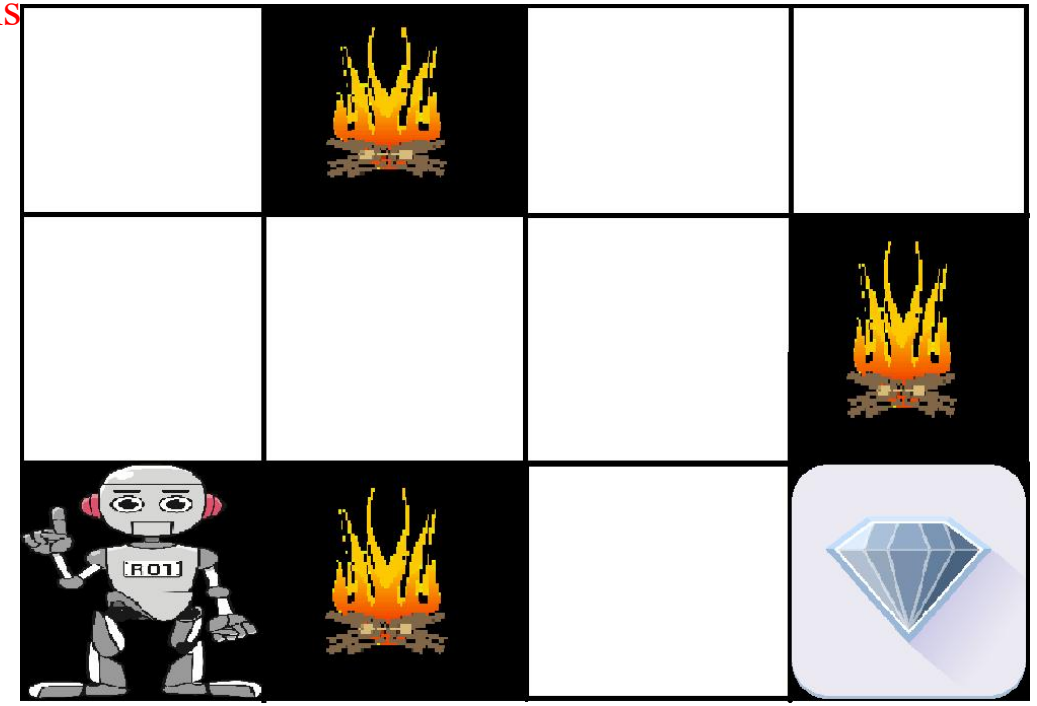


Deep neural network

Introduction

- Reinforcement learning
  - From different actions and states to learn best actions

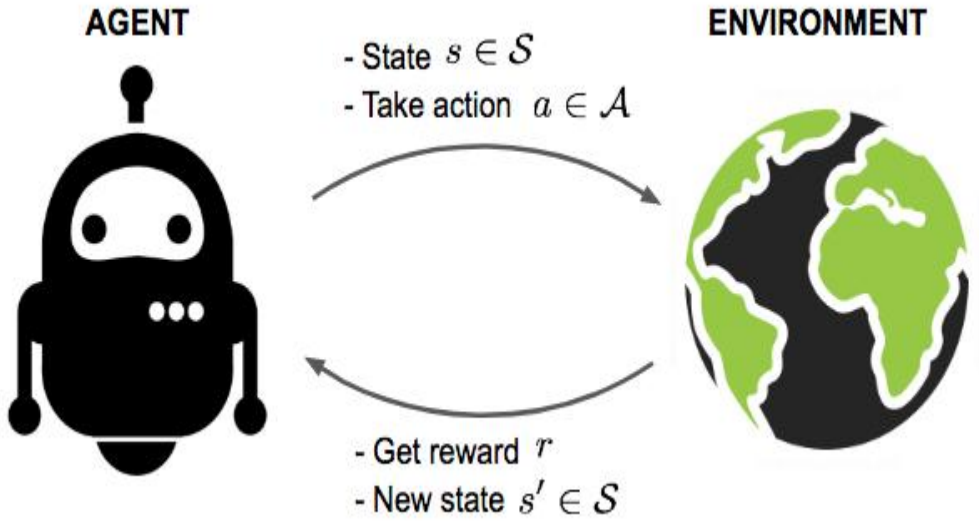


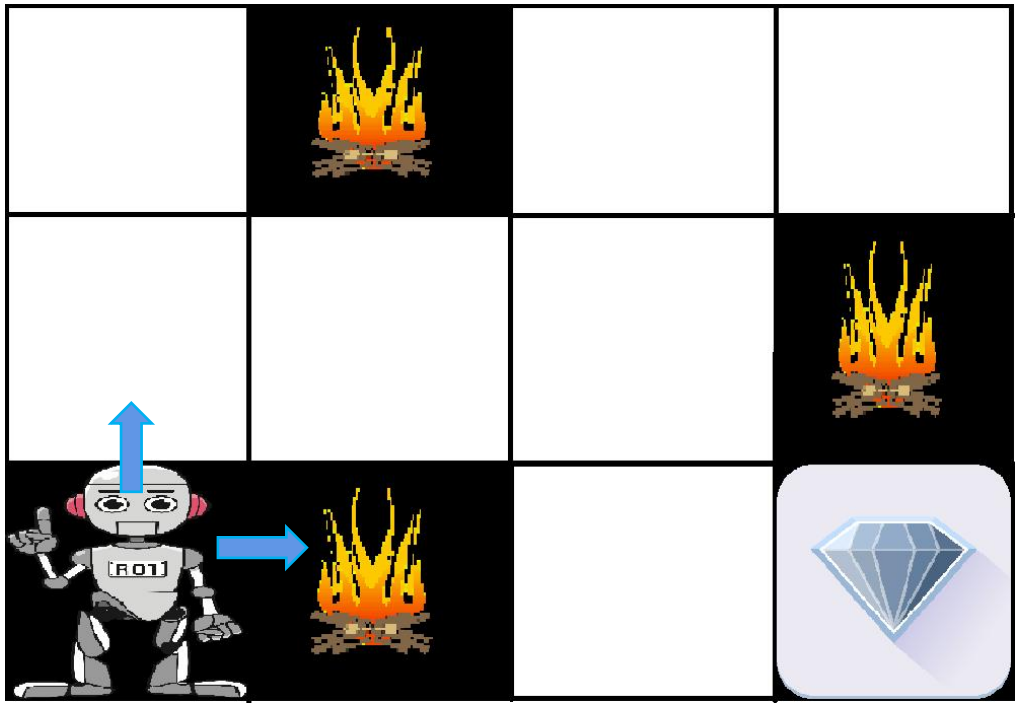


Training: Indicate the best action and give it the highest probability

Introduction

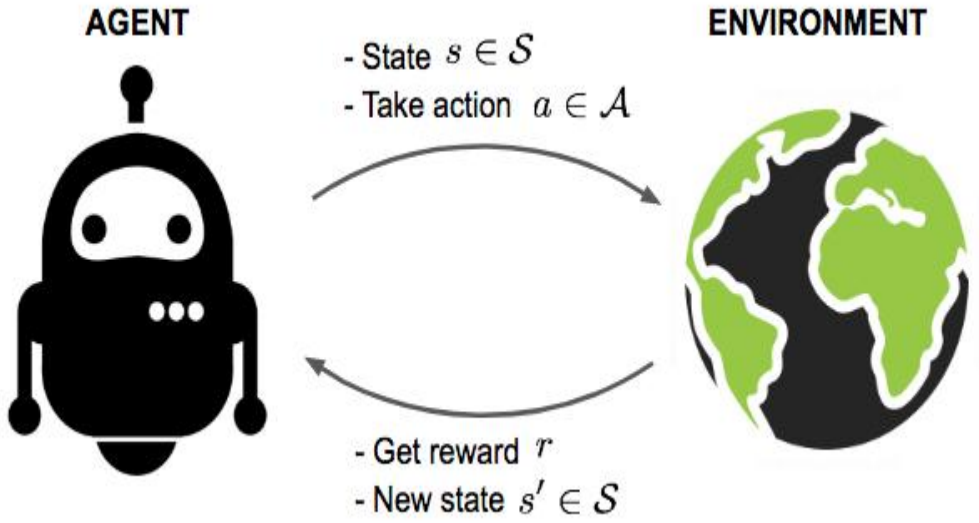
- Reinforcement learning





Introduction

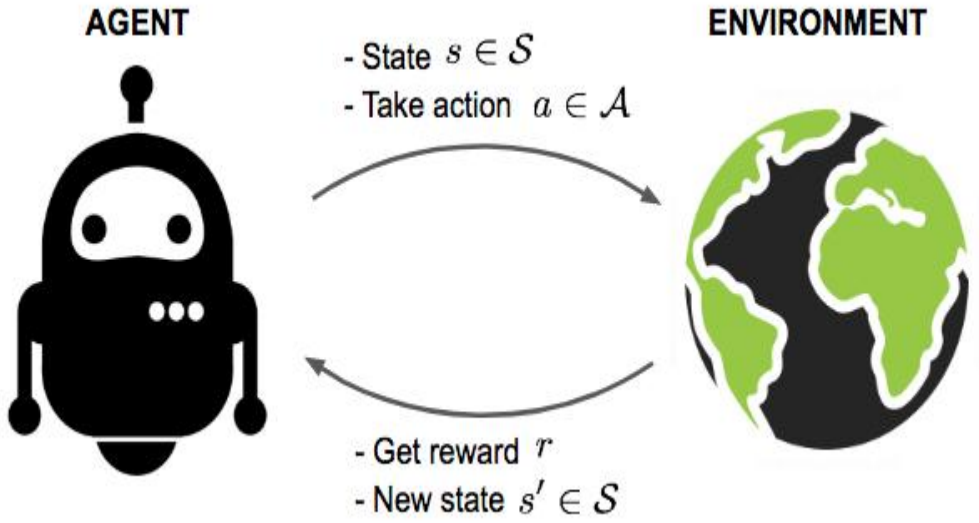
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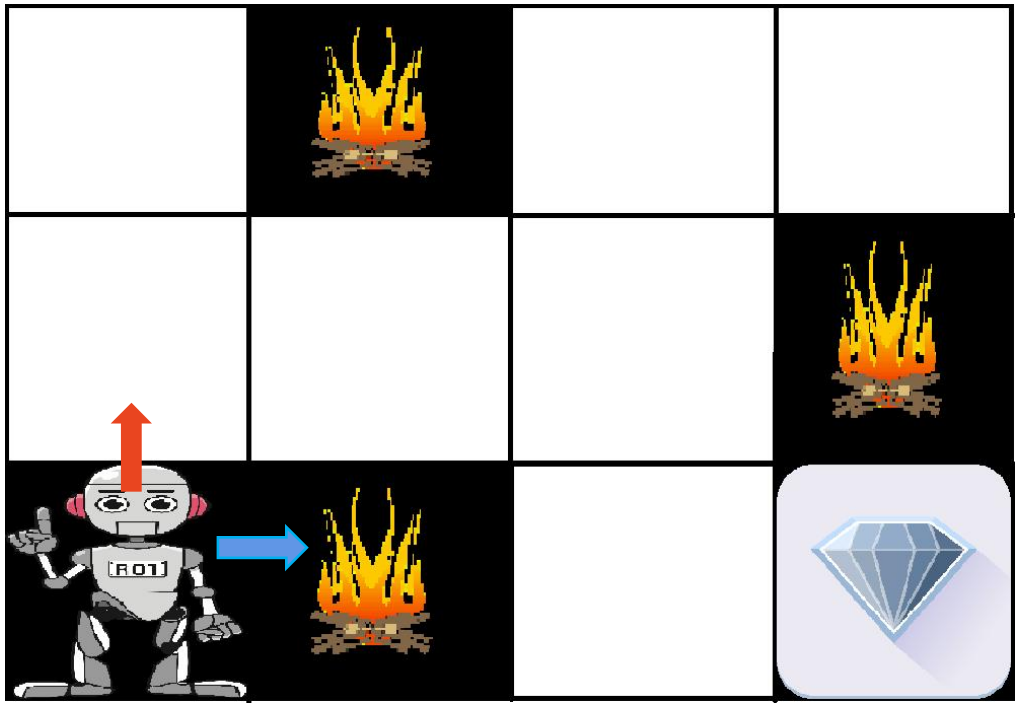




Introduction

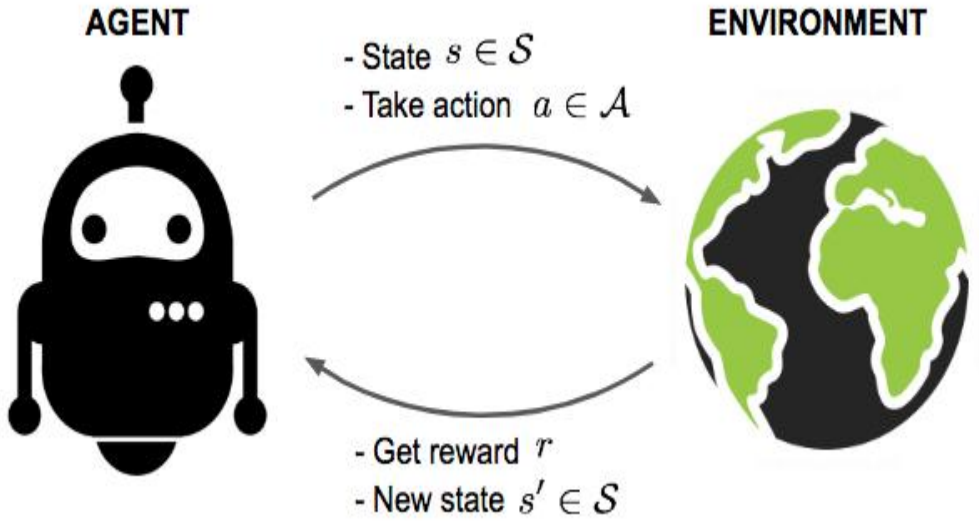
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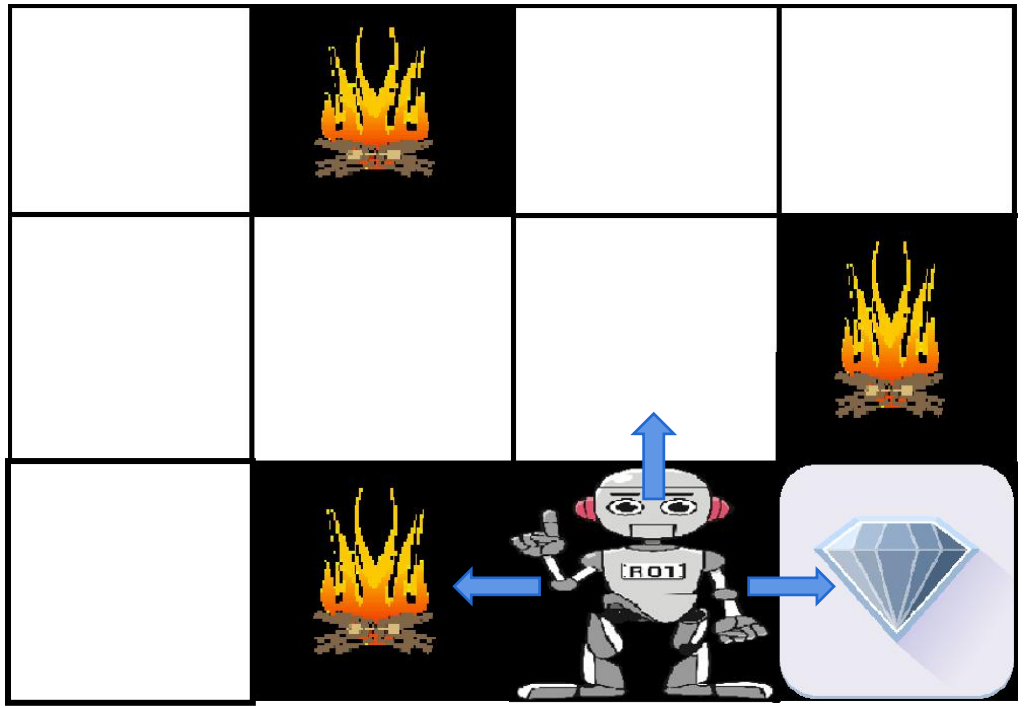




Introduction

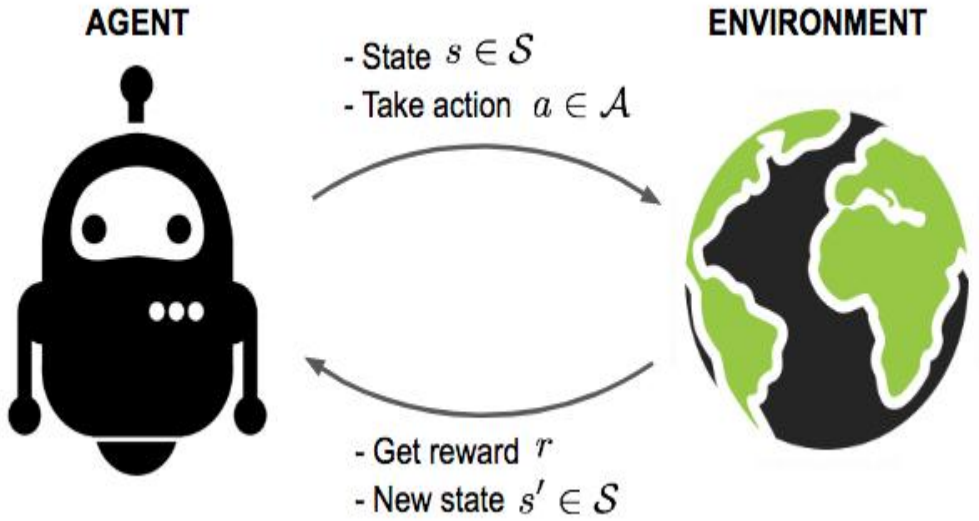
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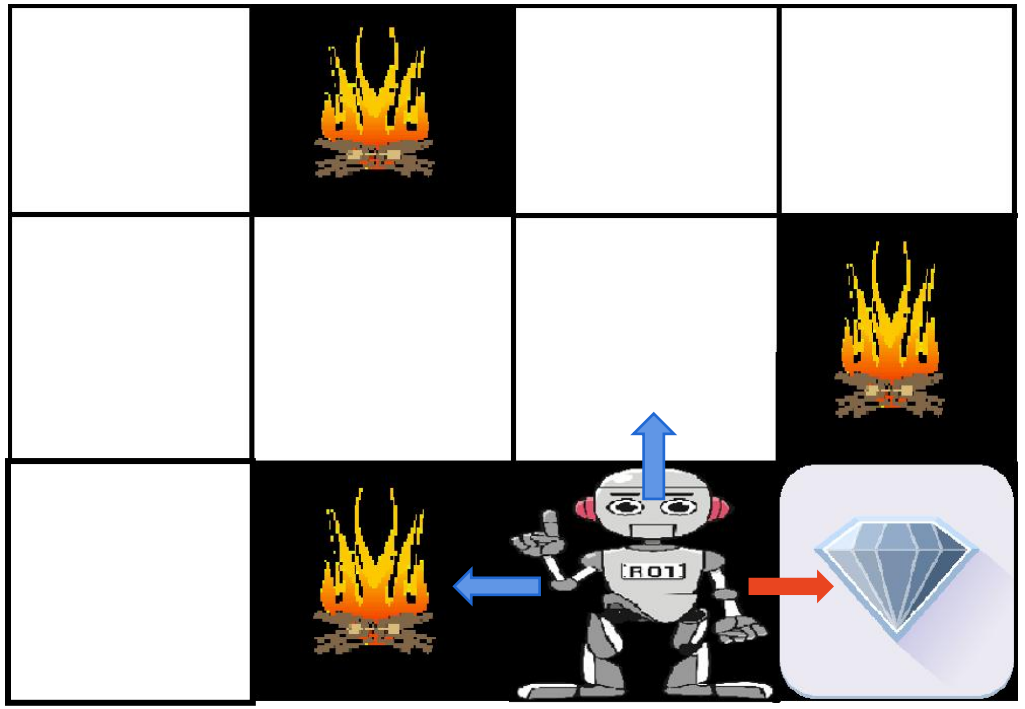




Introduction

- Reinforcement learning





Introduction

- Statistics and probability are in everywhere
- The cornerstone of almost every science

1. Population and sample
2. Pictorial and tabular method in descriptive statistics
3. Measure of location
4. Measure of variability

Population and sample

- Population
  - An investigation will typically focus on a well-defined collection of objects
    - Object is an individual unit of the investigation
    - A student in a school, a car made in a factory
  - Population is the set of all objects
    - All students in a school, all cars made in a factory

- Variable
  - Characteristic or charicteristics of each object in the population
    - Age of each student
    - Length of each car

Population and sample

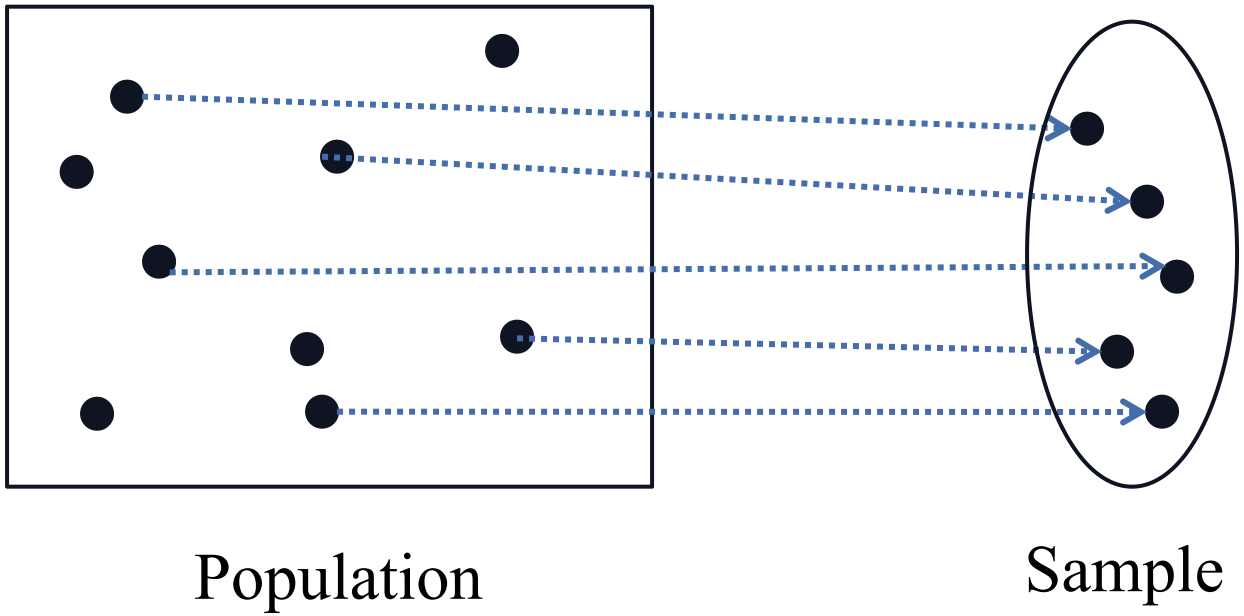
- Type of Variable
  - Univariate
    - A single value, e.g., age of student
  - Bivariate
    - Two values, e.g., height and weight of student
  - Multivariate
    - More than two values, e.g., age, height, and weight of student

Population and sample

Population	Object	Variable
All students in a class	Student	•Height •Weight •Hours of work per week
All circuit boards in a factory	Board	•Type of defects •Number of defects •Location of defects
All restaurants around a campus	Restaurant	•Number of employees •Number of seats
All books in a library	Book	•Cost •Frequency of checkout

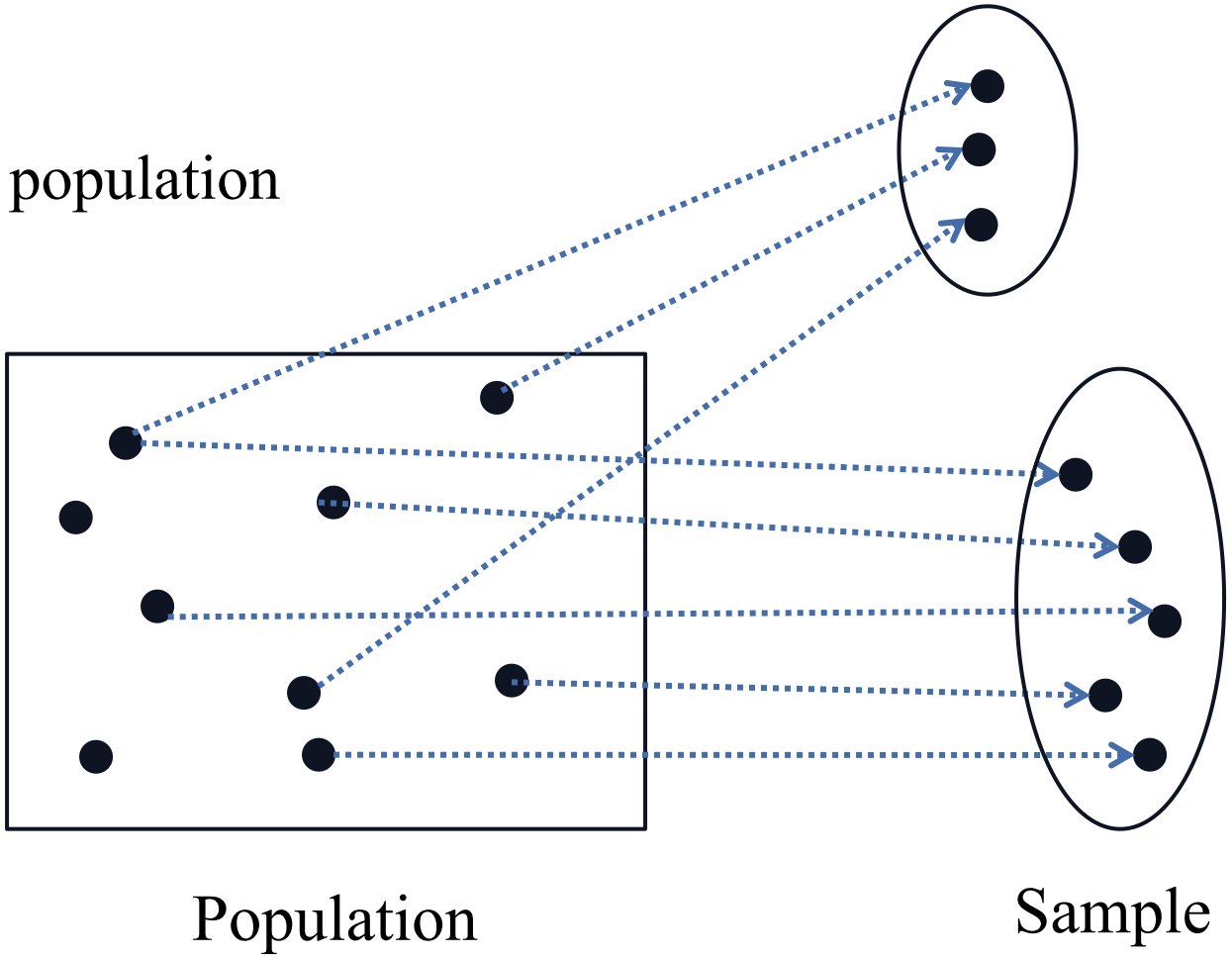
Population and sample

- Sample
  - A subset of population



Population and sample

- Sample
  - A subset of population



Population and sample

- Data collection
  - A sample is a subset of population, so we need collect data (objects) from population
  - Random sampling
    - Identical chance of selection
  - Stratified sampling
    - Systematic data selection

Population and sample

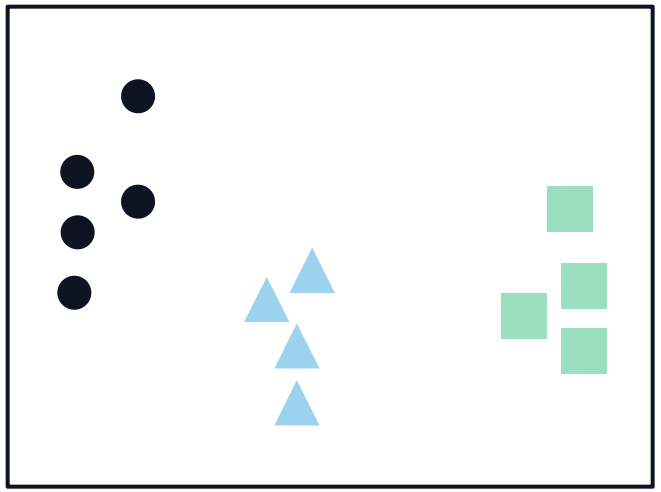
- Random sampling
  - Computer software
  - Mix and select one by one

Population and sample

- Random sampling
  - Computer software
  - Mix and select one by one
- Stratified sampling

Population and sample

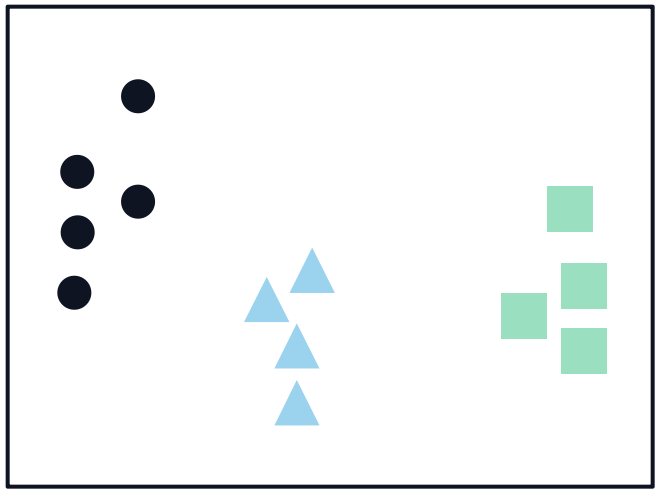
- Random sampling
  - Computer software
  - Mix and select one by one
- Stratified sampling
  - Distinct categories



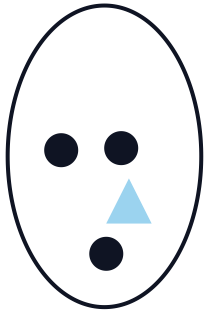
Population

Population and sample

- Random sampling
  - Computer software
  - Mix and select one by one
- Stratified sampling
  - Distinct categories

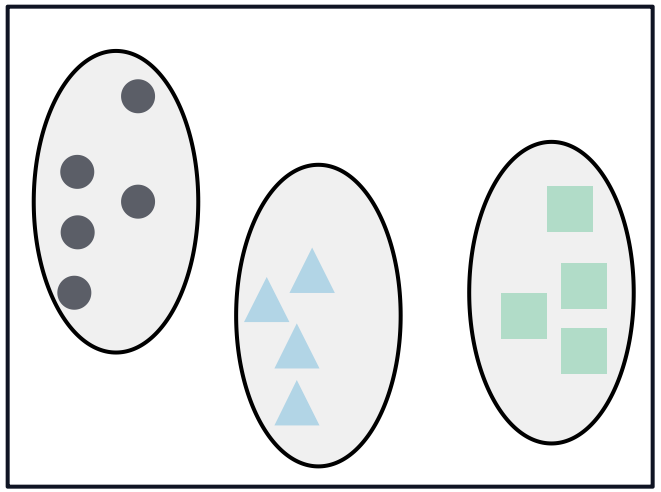


Population



Population and sample

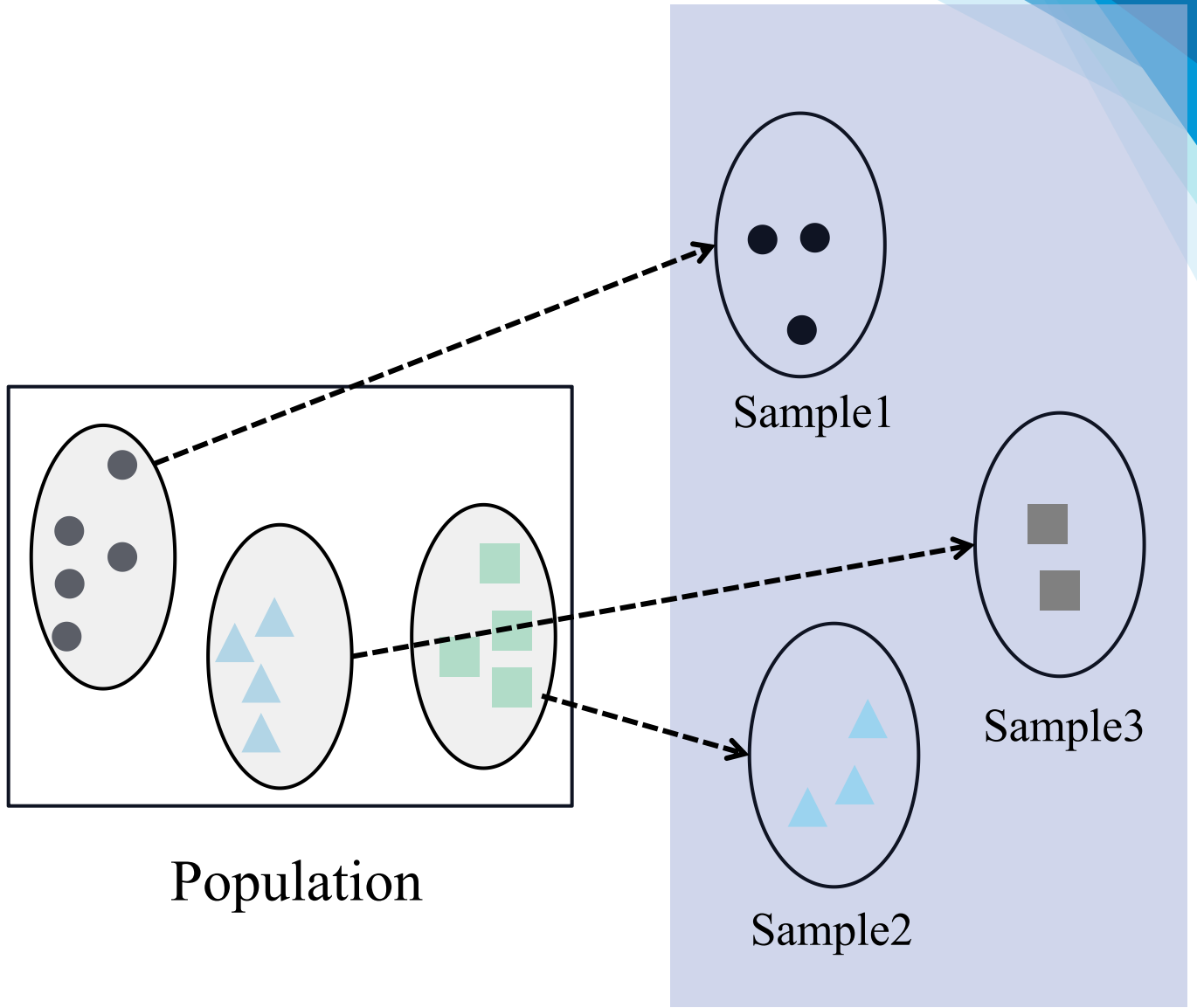
- Random sampling
  - Computer software
  - Mix and select one by one
- Stratified sampling
  - Distinct categories
  - Group each category



Population

Population and sample

- Random sampling
  - Computer software
  - Mix and select one by one
- Stratified sampling
  - Distinct categories
  - Group each category
  - Draw sample from each category



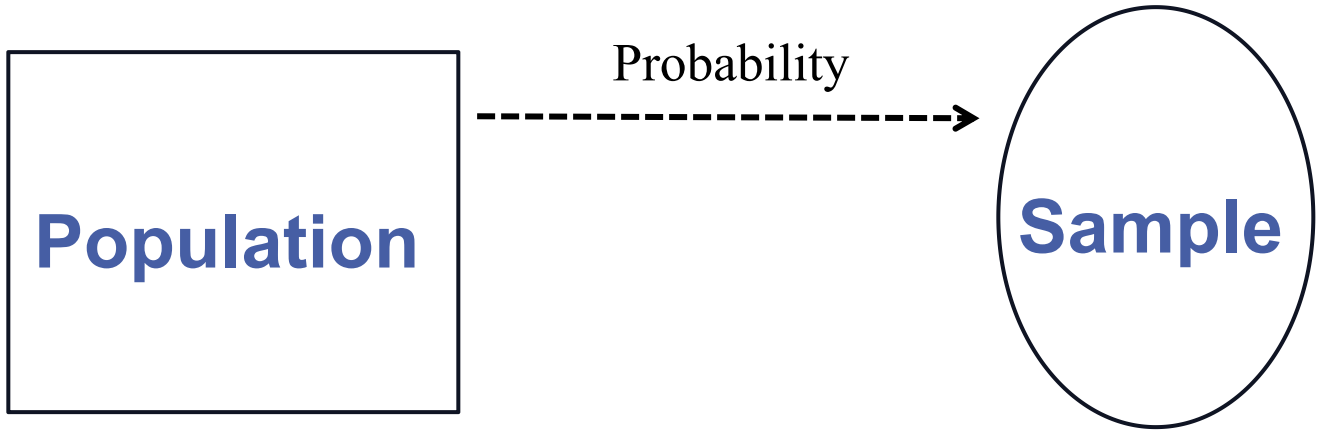
Population and sample

- The relationship between statistics and probability



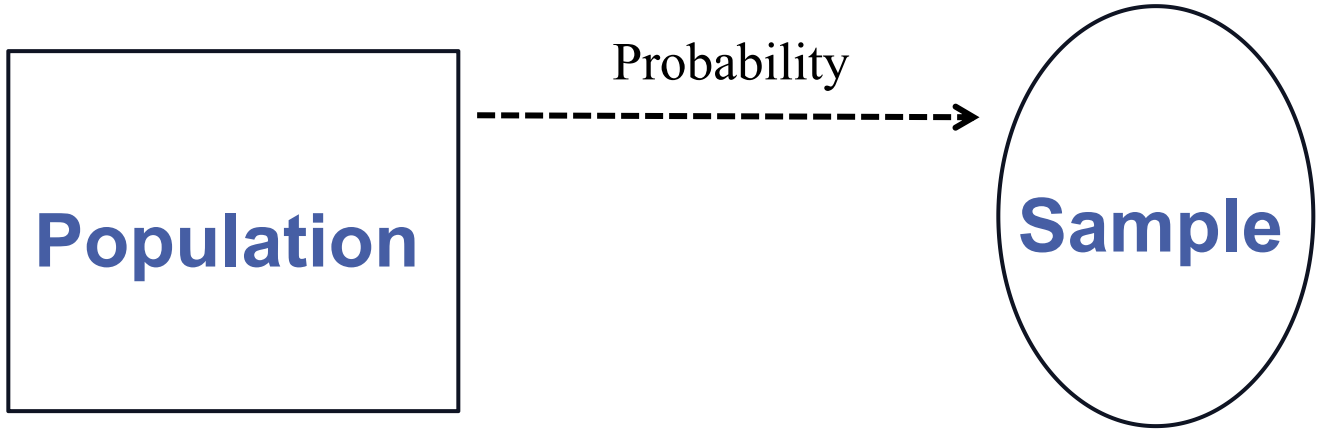
Population and sample

- The relationship between statistics and probability



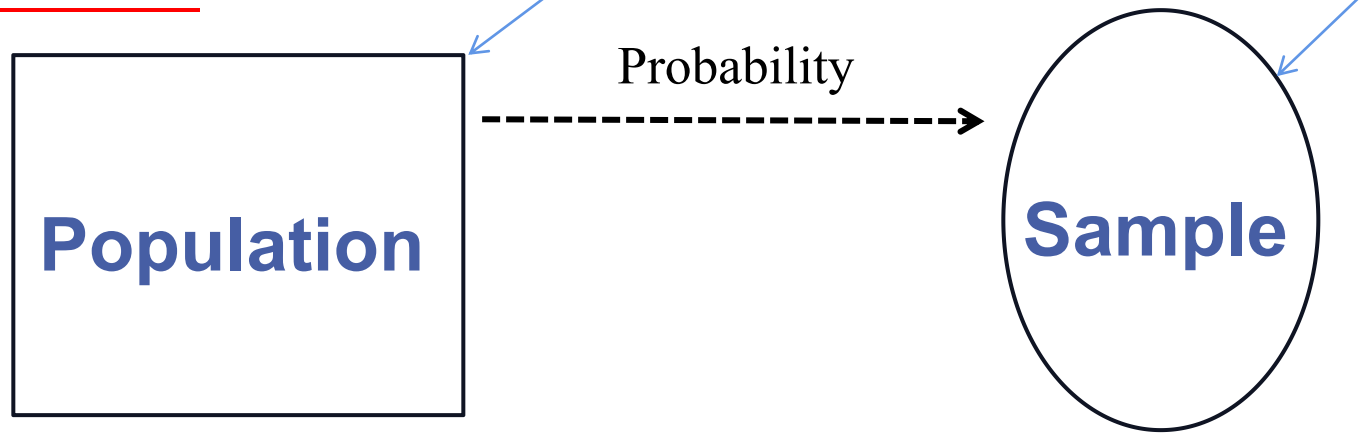
Population and sample

- The relationship between statistics and probability
  - The number of male students in a class of 50 students is 10
  - What is the probability for 5 male students to be selected if 30 students are selected randomly?



Population and sample

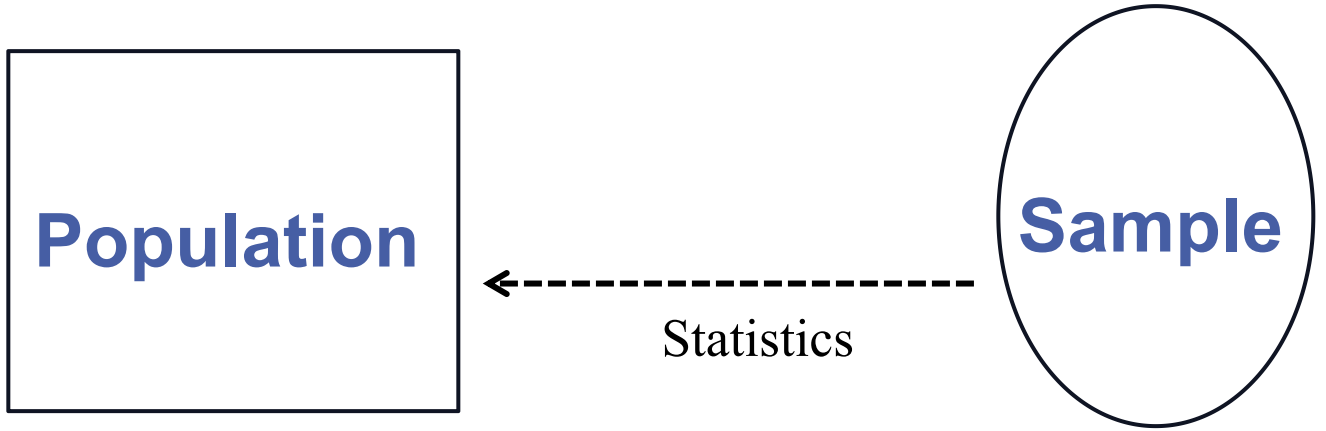
- The relationship between statistics and probability
  - The number of male students in a class of 50 students is 10
  - What is the probability for 5 male students to be selected if 30 students are selected randomly?



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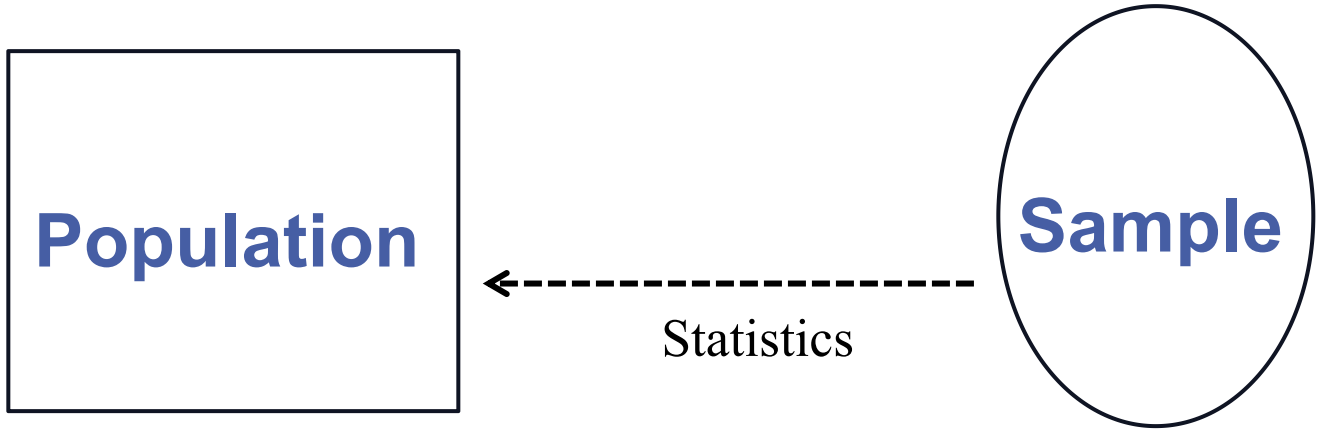
Population and sample

- The relationship between statistics and probability



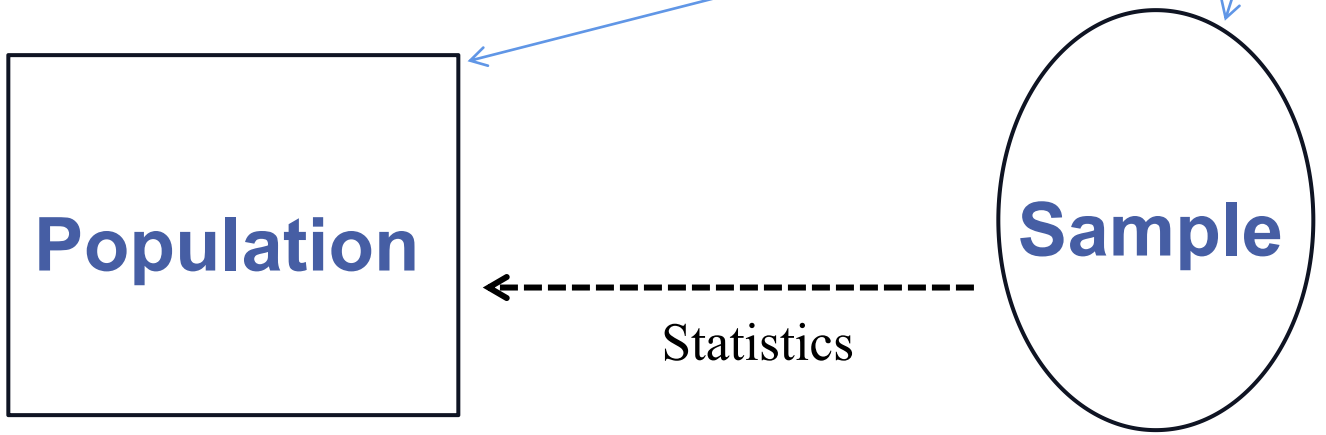
Population and sample

- The relationship between statistics and probability
  - There are 3 male students among 30 randomly selected students
  - How many male students can be there in the entire class?



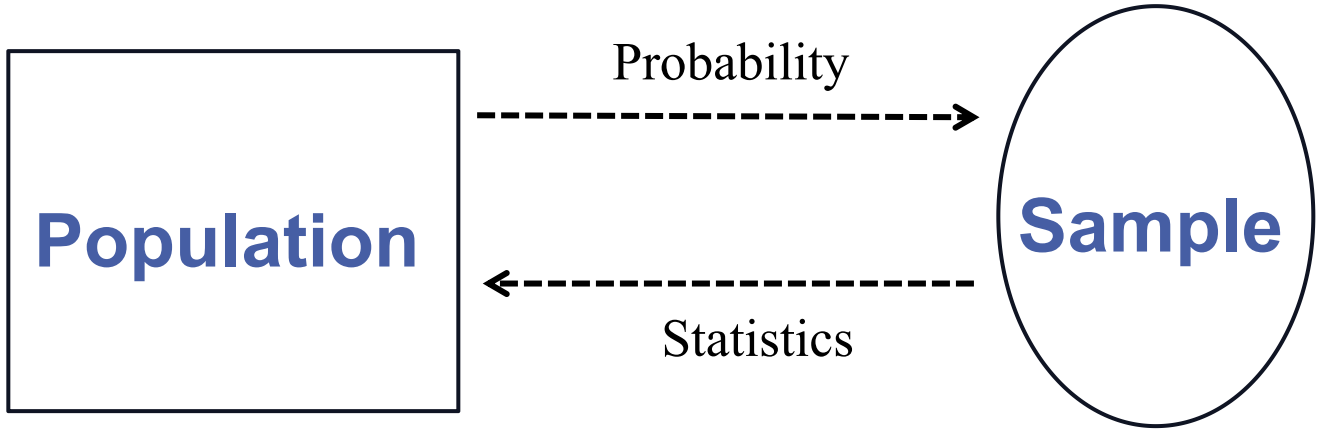
Population and sample

- The relationship between statistics and probability
  - There are 3 male students among 30 randomly selected students
  - How many male students can be there in the entire class?



Population and sample

- The relationship between statistics and probability
  - They are mutually opposite concepts



Population and sample

- Descriptive statistics

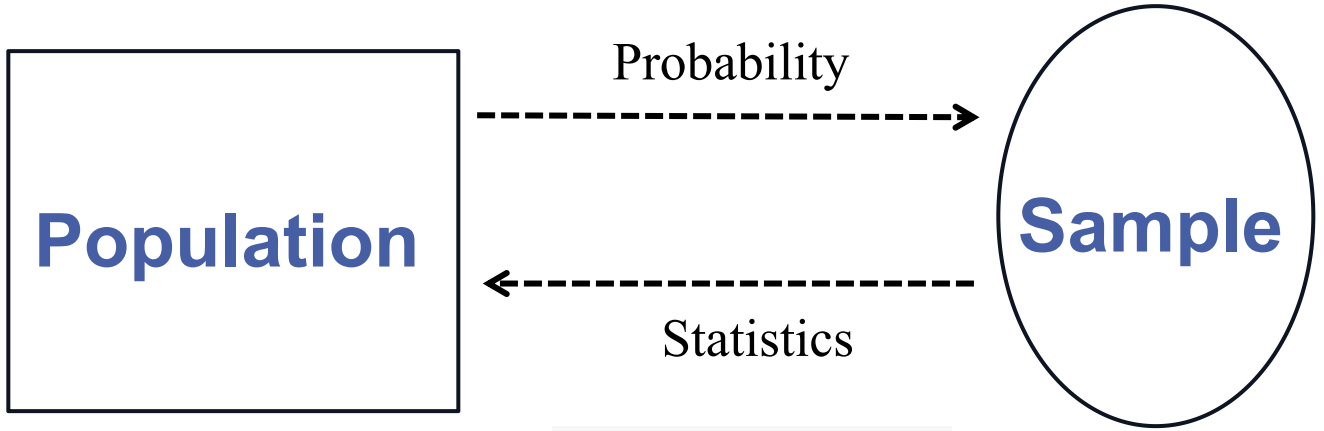
- Inferential statistics

Population and sample

- Descriptive statistics
  - To summarize and describe a sample
    - Graphical method
      - Steam-and-leaf, dotplot, histogram
    - Numerical method
      - Measure of location, measure of variability
- Inferential statistics
  - To draw conclusions about the population from the sample
    - Point Estimation
    - Hypothesis testing
    - Estimation by confidence interval

Population and sample

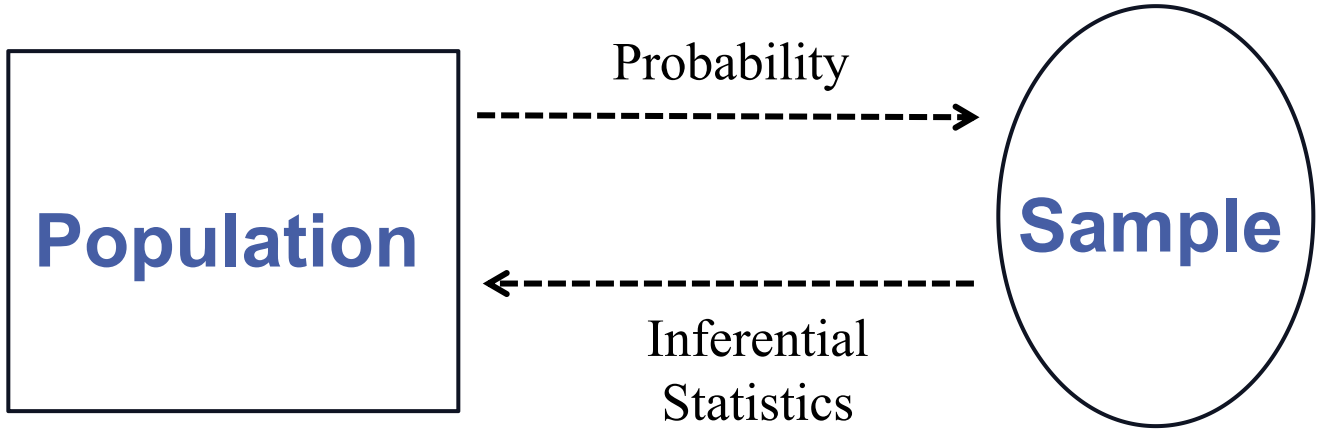
- The relationship between statistics and probability





Population and sample

- The relationship between statistics and probability



1. Population, sample, and process

2. Pictorial and tabular method in descriptive statistics

3. Measure of location

4. Measure of variability

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Pictorial and tabular method in descriptive statistics

- Descriptive statistics
  - To summarize and describe features of sample
    - Graphical method - visual information
      - Steam-and-leaf, dotplot, histogram
    - Numerical method - numerical information
      - Measure of location, measure of variability

Pictorial and tabular method in descriptive statistics

- Stem-and-leaf

Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31

Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31
  - Select one or more leading digits for the stems,
  - Trailing digits become the leaves
    - 16, 33, 64, 37, 31

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Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31
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Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31
  - Select one or more leading digits for the stem values,
  - Trailing digits become the leaves
    - 16, 33, 64, 37, 31
  - List stems

Stem	Leaf
1	
3	
6	

Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31
  - Select one or more leading digits for the stem values, trailing digits become the leaves
    - 16, 33, 64, 37, 31
  - List stems
  - List leaves

Stem	Leaf
1	6
3	3 7 1
6	4

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Pictorial and tabular method in descriptive statistics

- Stem-and-leaf
  - Suppose we have sample data
    - 16, 33, 64, 37, 31
  - Select one or more leading digits for the stem values, trailing digits become the leaves
    - 16, 33, 64, 37, 31
  - List stems
  - List leaves
  - Show units of stem and leaf

Stem	Leaf	
1	6	
3	3 7 1	Stem:tens digit Leaf:ones digit
6	4	

Pictorial and tabular method in descriptive statistics

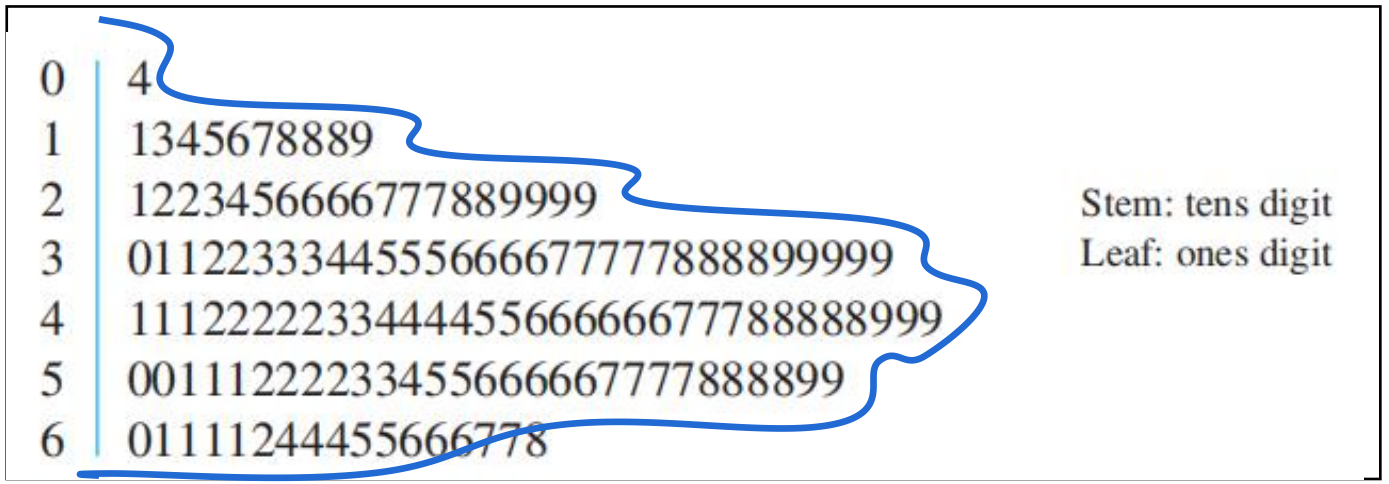
- Stem-and-leaf example
  - Sample of binge drinker percentages in 140 colleges

0	4
1	1345678889
2	12234566667778899999
3	01122333445556666777778888999999
4	11122222334444556666667778888999
5	00111222233455666667777888899
6	0111124445566677 8

Stem: tens digit
Leaf: ones digit

Pictorial and tabular method in descriptive statistics

- Stem-and-leaf example
  - Sample of binge drinker percentages in 140 colleges



0	4
1	1345678889
2	12234566667778899999
3	01122333445556666777778888999999
4	11122222334444556666667778888899
5	00111222233455666667777888899
6	0111124445566677 8

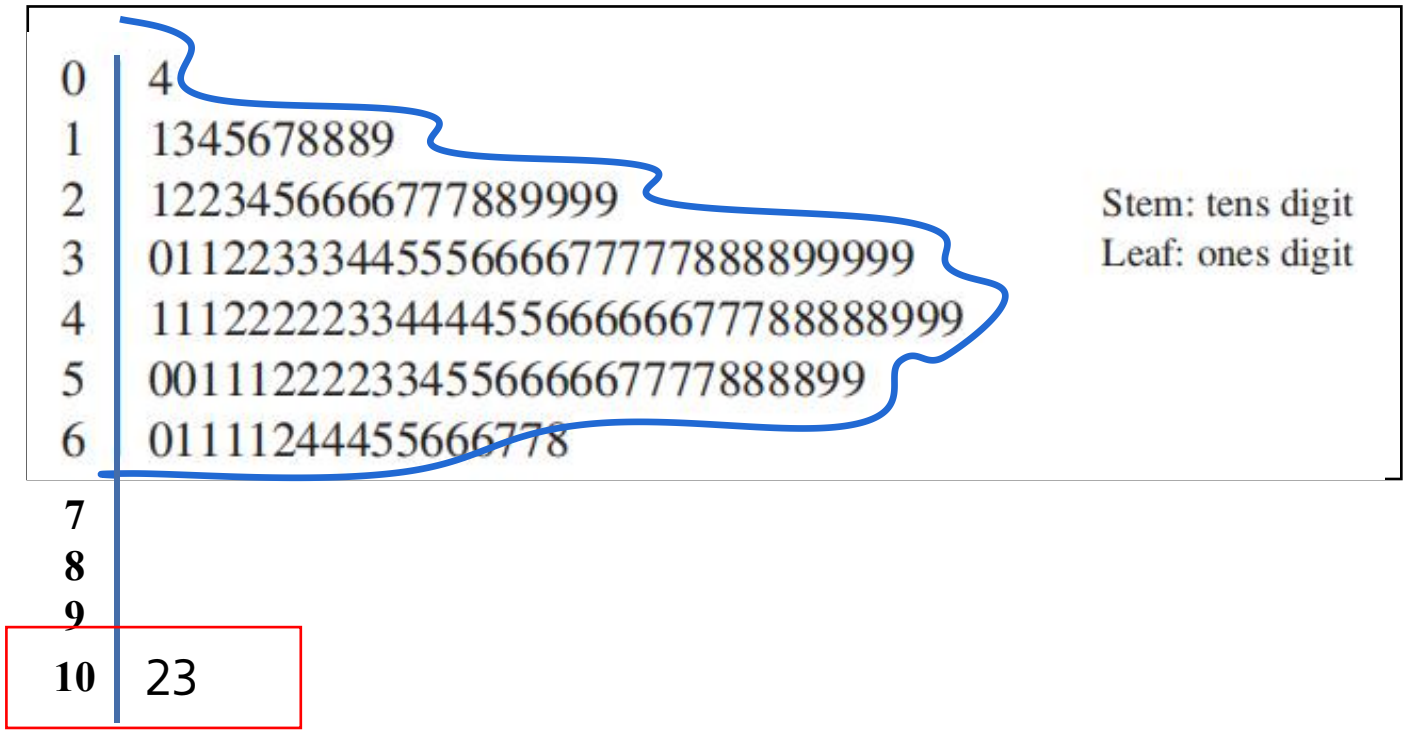
Stem: tens digit
Leaf: ones digit

  - The first leaf of stem 5 row is 0, so 50% of students in a college are binge drinker
  - Concentration: within the 40-49% range
  - Shape: stretching more towards the lower values
  - No gap nor outlier

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Pictorial and tabular method in descriptive statistics

- Stem-and-leaf example
  - Sample of binge drinker percentages in 140 colleges



Pictorial and tabular method in descriptive statistics

- Dotplot
  - Each value is represented by a dot
  - When a value occurs more than once, there is a dot for each occurrence, and these dots are stacked vertically

Pictorial and tabular method in descriptive statistics

- Dotplot example
  - Sample of O-ring temparature (F) of rocket engine

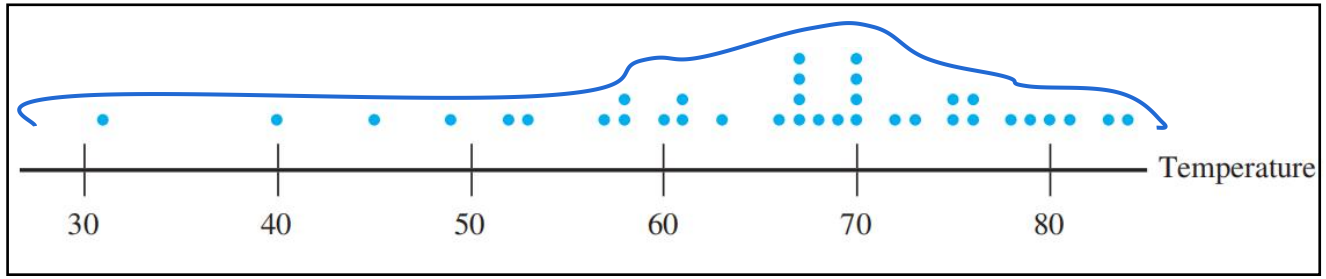
84	49	61	40	83	67	45	66	70	69	80	58
68	60	67	72	73	70	57	63	70	78	52	67
53	67	75	61	70	81	76	79	75	76	58	31

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Pictorial and tabular method in descriptive statistics

- Dotplot example
  - Sample of O-ring temparature (F) of rocket engine

84	49	61	40	83	67	45	66	70	69	80	58
68	60	67	72	73	70	57	63	70	78	52	67
53	67	75	61	70	81	76	79	75	76	58	31

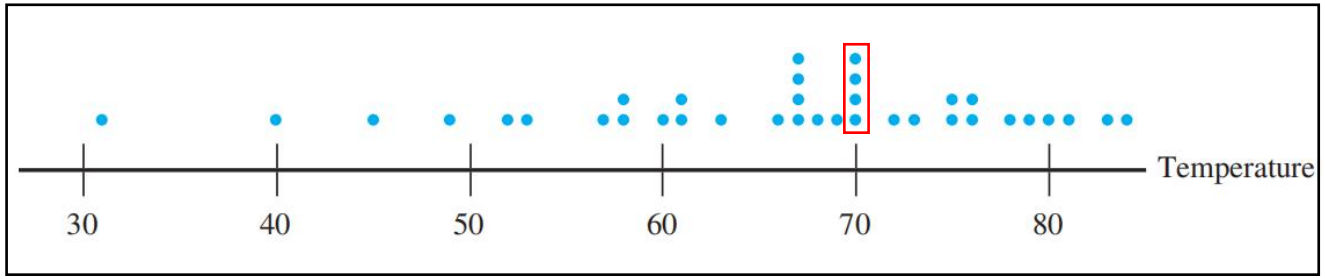


  - The data stretches out more at the lower end
  - The smallest observation, 31, is an outlier

Pictorial and tabular method in descriptive statistics

- Dotplot example
  - Sample of O-ring temparature (F) of rocket engine

84	49	61	40	83	67	45	66	70	69	80	58
68	60	67	72	73	70	57	63	70	78	52	67
53	67	75	61	70	81	76	79	75	76	58	31



Pictorial and tabular method in descriptive statistics

- Histogram
  - Type of variable
    - Discrete
      - Its set of possbile values is either finite or can be listed sequentially, e.g., natural number, integer
      - Number of people in a bus
    - Continuous
      - If its set of possbile values consists of an entire interval of a number line, e.g., real number
      - Reaction time to a stimulus

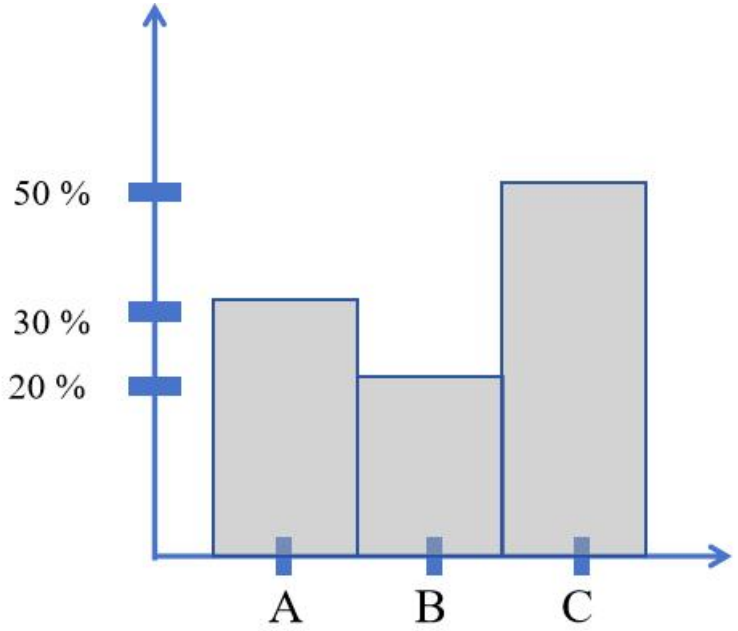
Pictorial and tabular method in descriptive statistics

- Histogram shows frequency or relative frequency
  - Frequency
    - Number of times a value occurs in a data set
    - If the number of students with 3.0 GPA among 60 students is 5, then frequency is 5
  - Relative Frequency
    - Fraction or proportion of times a value occurs
    - Relative frequency in above example is 5/60 = 8%

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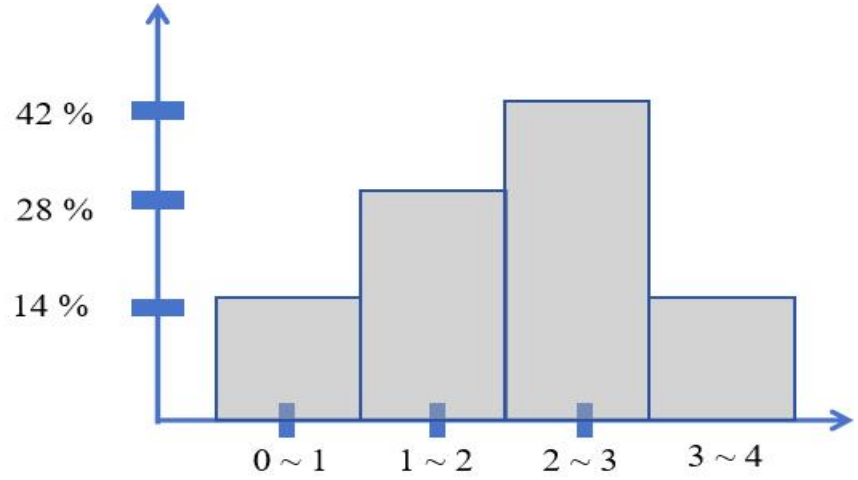
Pictorial and tabular method in descriptive statistics

- Histogram with discrete variable
  - Suppose we have sample data
    - A, A, A, B, B, C, C, C, C, C
  - Determine frequency or relative frequency of each value
    - A: 3 / 10 = 30%, B: 2 / 10 = 20%, C: 5 / 10 = 50%
  - Draw a rectangle whose height is the frequency or relative frequency



Pictorial and tabular method in descriptive statistics

- Histogram with continuous variable
  - Suppose we have sample data
    - 0.7, 1.5, 1.6, 2.3, 2.5, 2.6, 3.1
  - Divide the data set into a number of classes
    - 0 ~ 1, 1 ~ 2, 2 ~ 3, 3 ~ 4
  - Determine frequency or relative frequency of each class
    - 0 ~ 1: 1 / 7 = 14%, 1 ~ 2: 2 / 7 = 28%, 2 ~ 3: 3 / 7 = 42%, 3 ~ 4: 1 / 7 = 14%
  - Draw rectangle



1. Population, sample, and process

2. Pictorial and tabular method in descriptive statistics

3. Measure of location

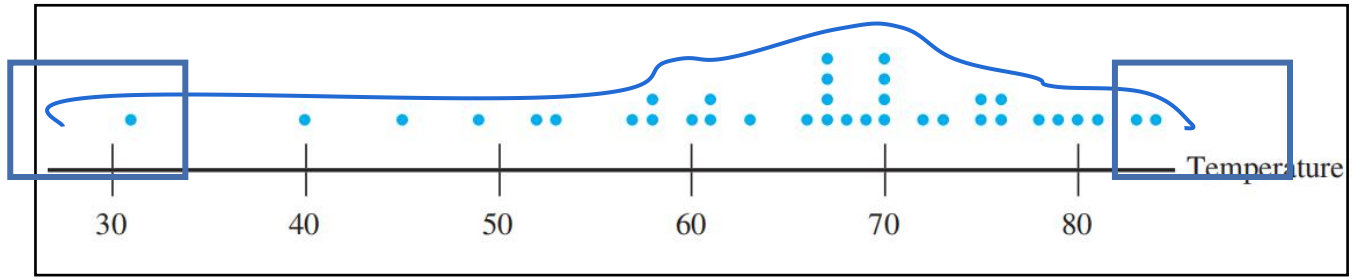
4. Measure of variability

Measure of location

- Descriptive statistics
  - To summarize and describe features of sample
    - Graphical method
      - Steam-and-leaf, dotplot, histogram
    - Numerical method
      - Measure of location, measure of variability
    - Although graphical method can provide preliminary impression and insight, we also need numerical method for the interpretation of numerical summary

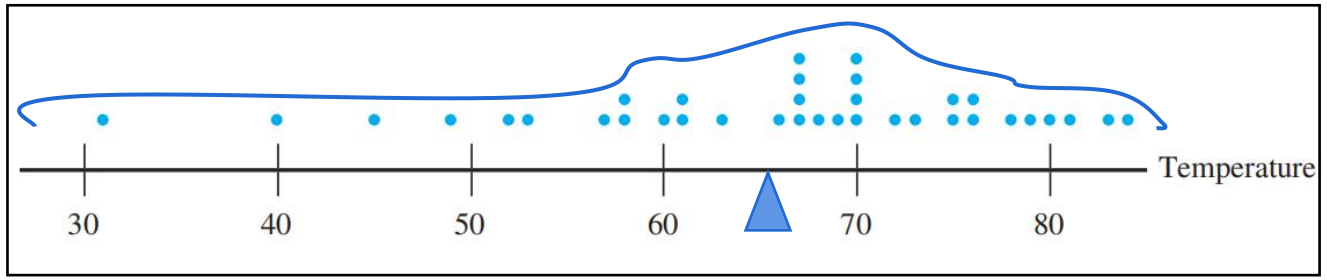
Measure of location

- Measure of location
  - Locational interpretation
  - Center location of a data set



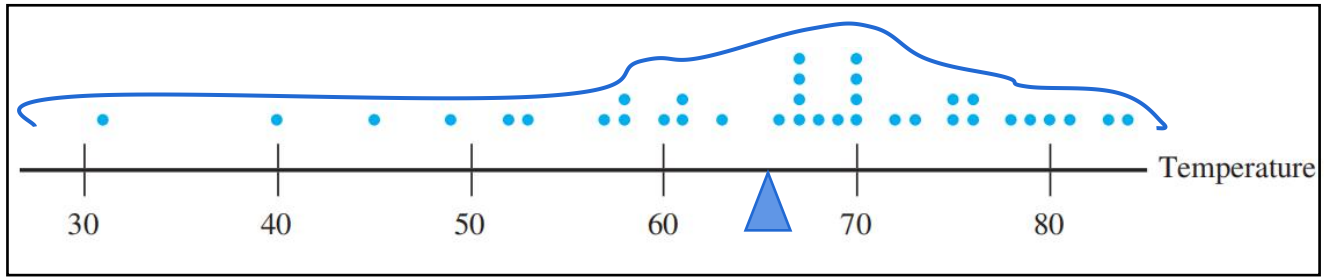
Measure of location

- Measure of location
  - Locational interpretation
  - Center location of a data set



Measure of location

- Measure of location
  - Locational interpretation
  - Center location of a data set
  - Mean, median



Measure of location

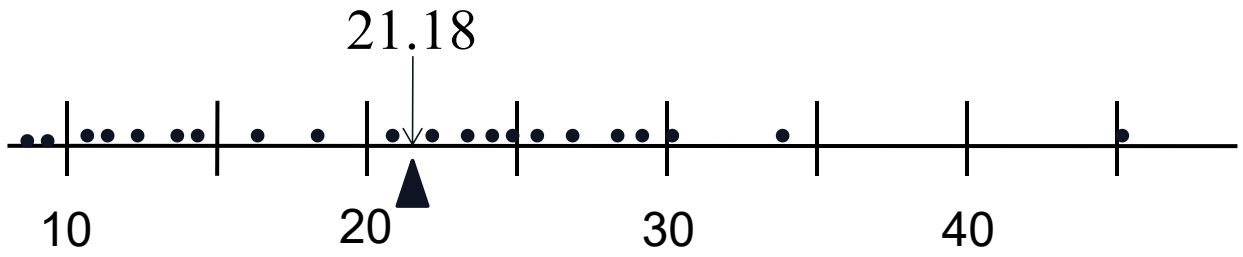
- Sample mean
  - The sample mean, usually denoted by $\bar{x}$, of observations $x_1, x_2, \ldots, x_n$ is given by

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

  - Sample mean can stand for weight balace point of data set

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$ $x_6 = 21.2$ $x_7 = 30.2$
$x_8 = 25.8$ $x_9 = 18.5$ $x_{10} = 10.3$ $x_{11} = 25.3$ $x_{12} = 14.0$ $x_{13} = 27.1$ $x_{14} = 45.0$
$x_{15} = 23.3$ $x_{16} = 24.2$ $x_{17} = 14.6$ $x_{18} = 8.9$ $x_{19} = 32.4$ $x_{20} = 11.8$ $x_{21} = 28.5$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 21.28$$



Measure of location

- Population mean
  - If there are *N* objects in population, the population mean, usually denoted by μ, is give by

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

Measure of location

- Sample median
  - Usaully denoted by $\tilde{x}$, the middle value once the observations are ordered from smallest to largest

$$\tilde{x} = \begin{cases} \text{The single middle value if } n \text{ is odd} & = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ ordered value} \\ \text{The average of the two middle values if } n \text{ is even} & = \text{average of } \left(\frac{n}{2}\right)^{\text{th}} \text{ and } \left(\frac{n}{2}+1\right)^{\text{th}} \text{ ordered values} \end{cases}$$

7.6 8.3 9.3 9.4 9.4 9.7 10.4 11.5 11.9 15.2 16.2 20.4 → $\tilde{x} = \frac{9.7 + 10.4}{2} = 10.05$

- Population median
  - Usually denoted by $\tilde{\mu}$, the middle value of population

Measure of location

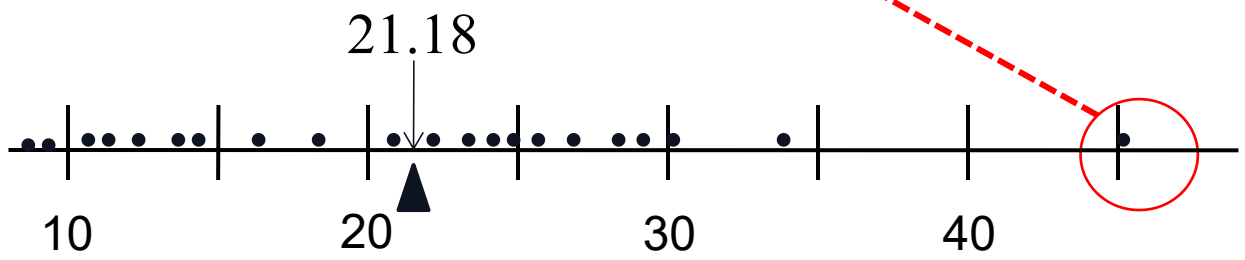
- Relationship between mean and median
  - Mean can be highly affected by an outlier

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$ $x_6 = 21.2$ $x_7 = 30.2$

$x_8 = 25.8$ $x_9 = 18.5$ $x_{10} = 10.3$ $x_{11} = 25.3$ $x_{12} = 14.0$ $x_{13} = 27.1$ $x_{14} = 45.0$

$x_{15} = 23.3$ $x_{16} = 24.2$ $x_{17} = 14.6$ $x_{18} = 8.9$ $x_{19} = 32.4$ $x_{20} = 11.8$ $x_{21} = 28.5$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 21.28$$

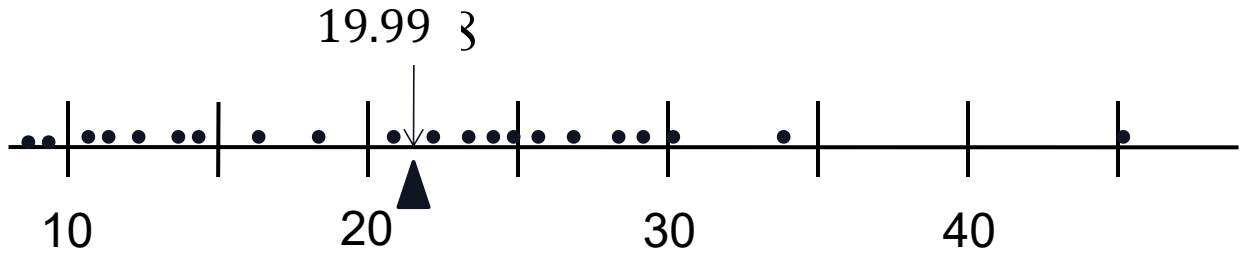


Measure of location

- Relationship between mean and median
  - Mean can be highly affected by an outlier

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$ $x_6 = 21.2$ $x_7 = 30.2$
$x_8 = 25.8$ $x_9 = 18.5$ $x_{10} = 10.3$ $x_{11} = 25.3$ $x_{12} = 14.0$ $x_{13} = 27.1$ ~~$x_{14} = 45.0$~~
$x_{15} = 23.3$ $x_{16} = 24.2$ $x_{17} = 14.6$ $x_{18} = 8.9$ $x_{19} = 32.4$ $x_{20} = 11.8$ $x_{21} = 28.5$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 19.99$$



Measure of location

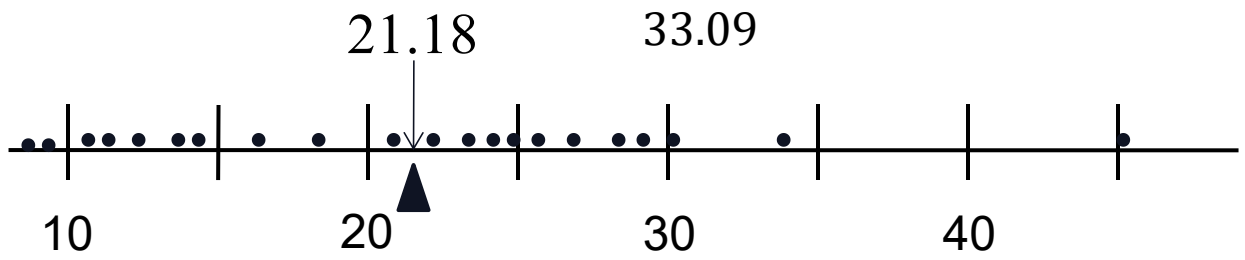
- Relationship between mean and median
  - Mean can be highly affected by an outlier

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$ $x_6 = 21.2$ $x_7 = 30.2$

$x_8 = 25.8$ $x_9 = 18.5$ $x_{10} = 10.3$ $x_{11} = 25.3$ $x_{12} = 14.0$ $x_{13} = 27.1$ $x_{14} = 295.0$

$x_{15} = 23.3$ $x_{16} = 24.2$ $x_{17} = 14.6$ $x_{18} = 8.9$ $x_{19} = 32.4$ $x_{20} = 11.8$ $x_{21} = 28.5$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = 33.09$$

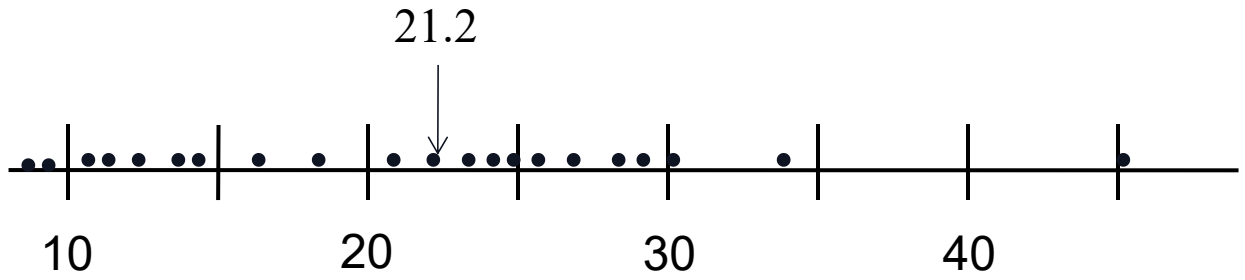


Measure of location

- Relationship between mean and median
  - Mean can be highly affected by an outlier
  - Median is insensible to outlier

$x_1 = 16.1$ $x_2 = 9.6$ $x_3 = 24.9$ $x_4 = 20.4$ $x_5 = 12.7$ $x_6 = 21.2$ $x_7 = 30.2$
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$\tilde{x} = 21.2$

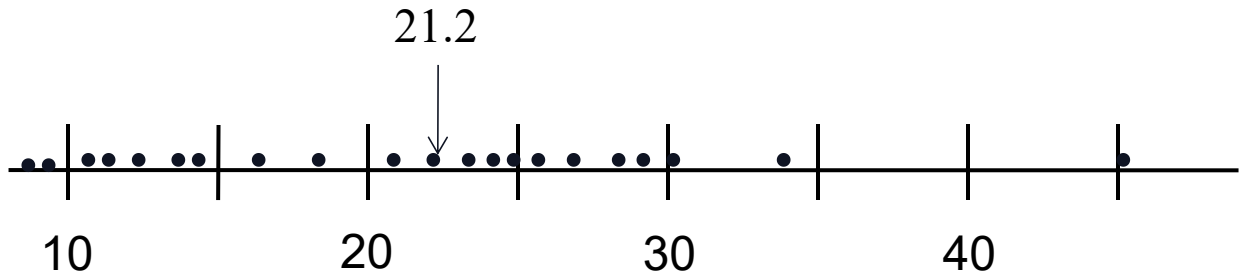


Measure of location

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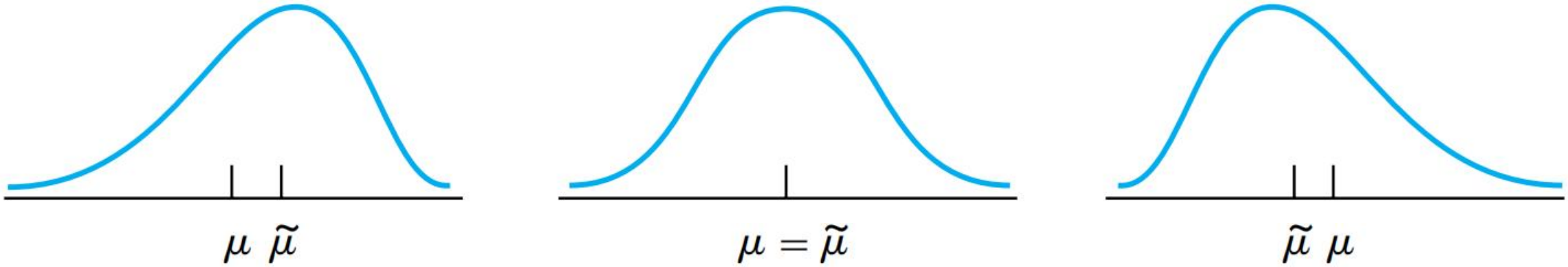
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$\tilde{x} = 21.2$



Measure of location

- Relationship between mean and median
  - How about if population mean and median are considered?
  - Most of time mean and median are not identical



Three distinct population histograms, negative skew, symmetric, positive skew

Measure of location

- Trimmed mean
  - A compromise between mean and median
  - Eliminate the smallest and the largest of the sample data set and then average over the left

Measure of location

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  - A compromise between mean and median
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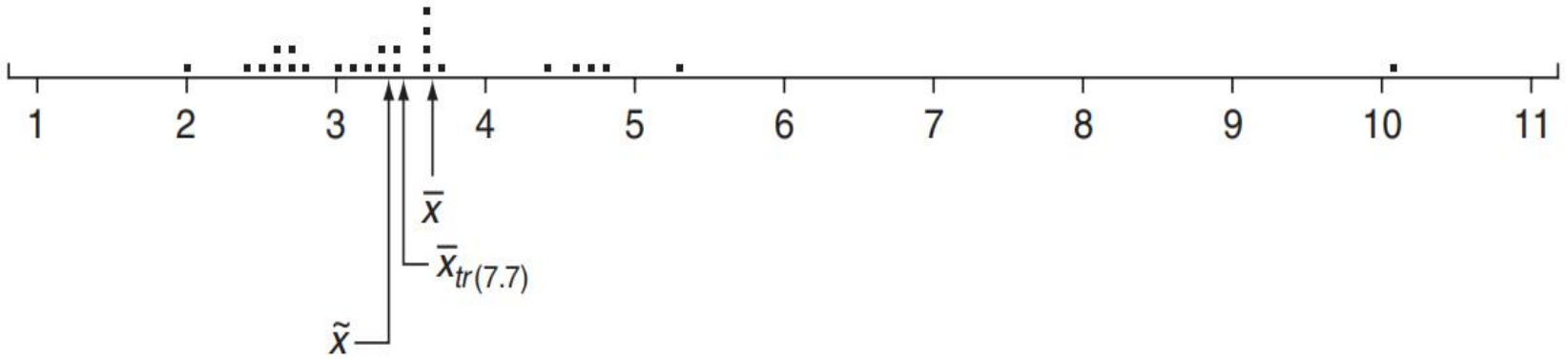
2.0	2.4	2.5	2.6	2.6	2.7	2.7	2.8	3.0	3.1	3.2	3.3	3.3
3.4	3.4	3.6	3.6	3.6	3.6	3.7	4.4	4.6	4.7	4.8	5.3	10.1

Measure of location

- Trimmed mean
  - A compromise between mean and median
  - Eliminate the smallest and the largest of the sample data set and then average over the left

~~2.0~~ ~~2.4~~ 2.5 2.6 2.6 2.7 2.7 2.8 3.0 3.1 3.2 3.3 3.3
3.4 3.4 3.6 3.6 3.6 3.6 3.7 4.4 4.6 4.7 4.8 ~~5.3~~ ~~10.1~~

  - Eliminate smallest two and largest two, $\frac{2}{26} = 7.7\%$, $\bar{x}_{tr(7.7)} = 3.42$



1. Population, sample, and process

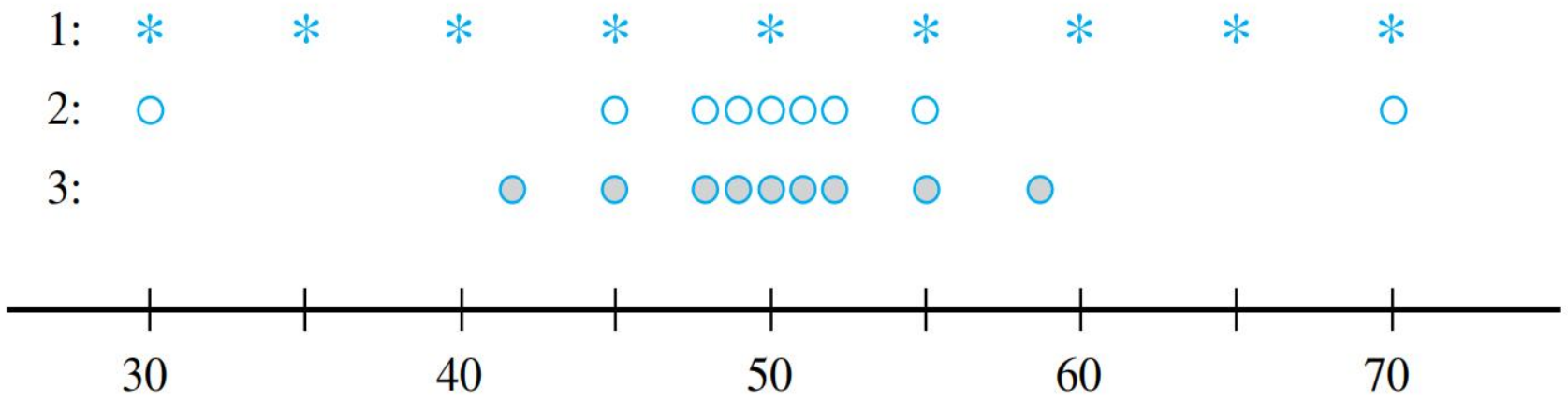
2. Pictorial and tabular method in descriptive statistics

3. Measure of location

4. Measure of variability

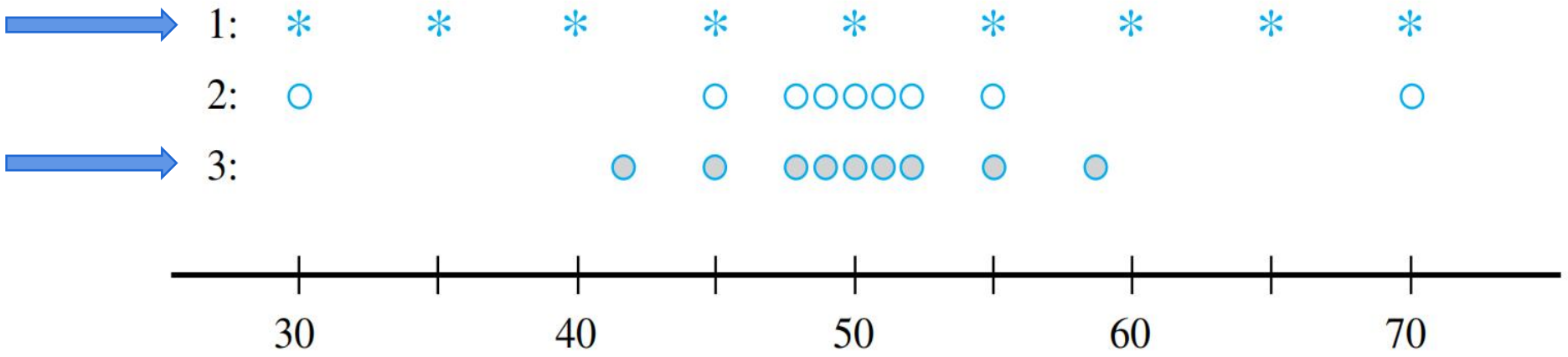
Measure of variability

- It is insufficient to only have center measure of data set
- Different samples may have identical center measure but distinct shapes
- Variability stands for the difference in data set



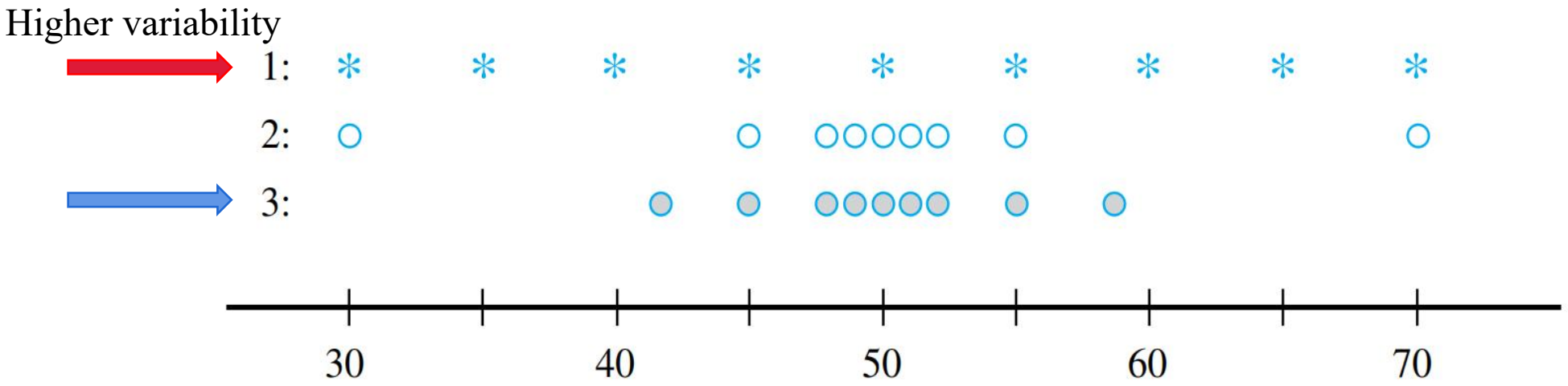
Measure of variability

- Range
  - Difference between largest value and smallest value



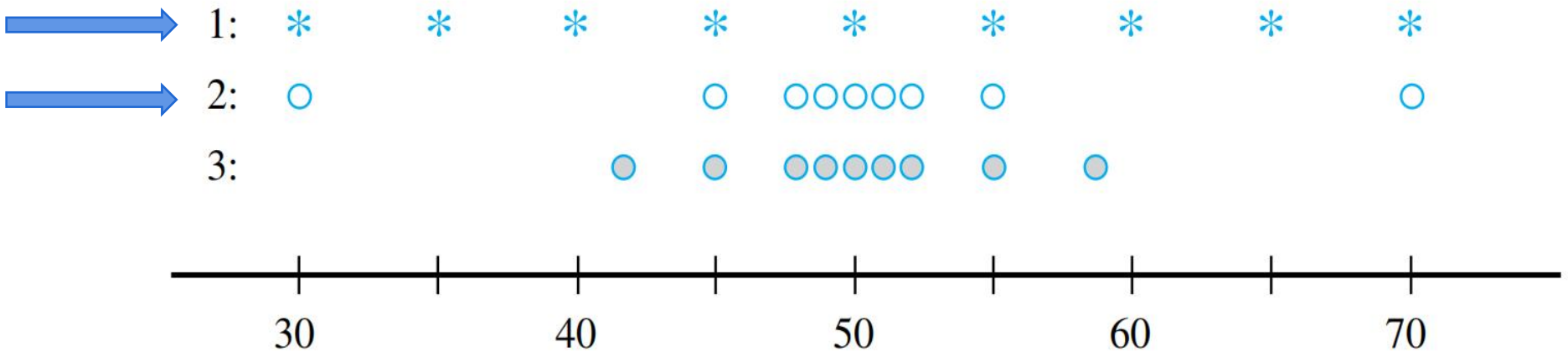
Measure of variability

- Range
  - Difference between largest value and smallest value



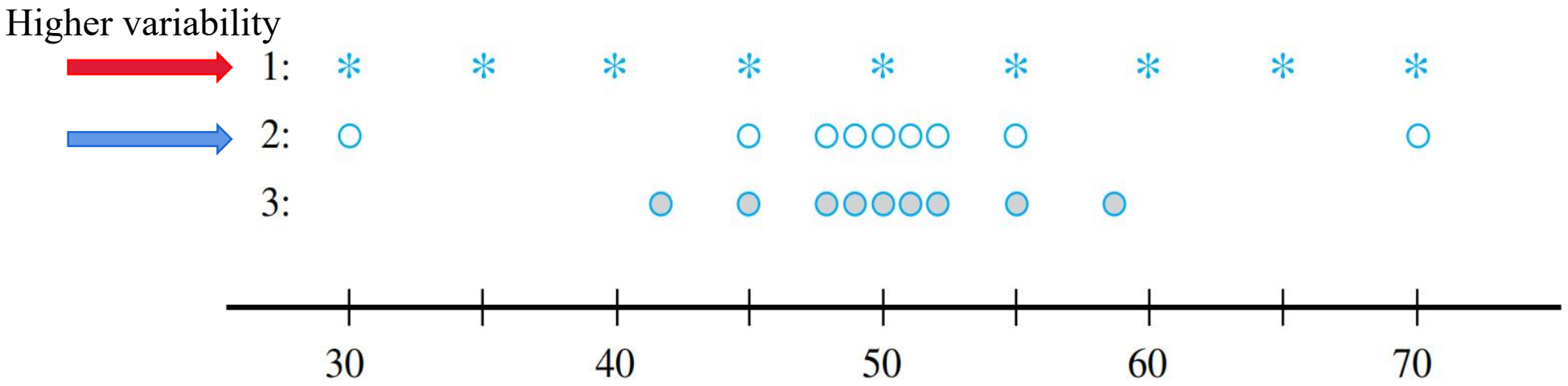
Measure of variability

- Range
  - Difference between largest value and smallest value



Measure of variability

- Range
  - Difference between largest value and smallest value
  - So only have range is not enough



Measure of variability

- Deviation
  - Subtracting mean from each data
  - $x_1 - \bar{x}, x_2 - \bar{x}, x_3 - \bar{x} \ldots x_n - \bar{x}$
  - If a deviation is large, it means big variability, vice versa
  - How to combine deviations into a single value?

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Measure of variability

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  - Subtracting mean from each data
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$$\text{sum of deviations} = \sum_{i=1}^{n} (x_i - \bar{x})$$

Measure of variability

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  - Subtracting mean from each data
  - $x_1 - \bar{x}, x_2 - \bar{x}, x_3 - \bar{x} \ldots x_n - \bar{x}$
  - If a deviation is large, it means big variability, vice versa
  - How to combine deviations into a single value?

$$\text{sum of deviations} = \sum_{i=1}^{n} (x_i - \bar{x})$$

$$\sum(x_i - \bar{x}) = \sum x_i - \sum \bar{x} = \sum x_i - n\bar{x} = \sum x_i - n\left(\frac{1}{n}\sum x_i\right) = 0$$ 

$$\sum_{i=1}^{n} (x_i - \bar{x})^2$$ 

Measure of variability

- Sample variance (S^2) and standard deviation (S)

$$\begin{cases} S^2 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1} \\ S = \sqrt{S^2} \end{cases}$$

- Population variance (σ^2) and standard deviation (σ)

$$\begin{cases} \sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \\ \sigma = \sqrt{\sigma^2} \end{cases}$$

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Measure of variability

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Not *n*
Bessel`s correction

- Population variance (σ^2) and standard deviation (σ)

$$\begin{cases} \sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \\ \sigma = \sqrt{\sigma^2} \end{cases}$$

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Measure of variability

- Sample variance (S^2) and standard deviation (S)

$$\begin{cases} S^2 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1} \\ S = \sqrt{S^2} \end{cases}$$

$$\sum (x_i - \bar{x})^2 < \sum (x_i - \mu)^2$$

Because generally x_is are closer to $\bar{x}$ than μ
The sample mean is dependent on x_is, but the population mean is fixed

- Population variance (σ^2) and standard deviation (σ)

$$\begin{cases} \sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \\ \sigma = \sqrt{\sigma^2} \end{cases}$$

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Measure of variability

- Computation of S^2

$$\sum (x_i - \bar{x})^2 = \sum \left(x_i^2 - 2\bar{x} \cdot x_i + \bar{x}^2\right) = \sum x_i^2 - 2\bar{x} \sum x_i + \sum \bar{x}^2$$

$$= \sum x_i^2 - 2\bar{x} \cdot n\bar{x} + n(\bar{x})^2 = \sum x_i^2 - n(\bar{x})^2$$

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Exercises

- Cumulative frequency and relative frequency
  - Sum of frequency or relative frequency of its own class and the classes preceding itself
  - If there are 4 intervals with frequencies 9, 16, 13, and 12,
    - What is the cumulative frequencies and cumulative relative frequencies of each class?

Exercises

- Cumulative frequency and relative frequency
  - Sum of frequency or relative frequency of its own class and the classes preceding itself
  - If there are 4 intervals with frequencies 9, 16, 13, and 12,
    - Cumulative frequency of each class is 9, 9+16 = 25, 9+16+13 = 38, and 9+16+13+12 = 50
    - Cumulative relative frequency of each class is 9 / 50 = 0.18, 25 / 50 = 0.5, 38 / 50 = 0.76, and 50/50 = 1.00

Exercises

- If we add a constant c to each of sample data, $y_1 = x_1 + c, y_2 = x_2 + c, y_3 = x_3 + c, \ldots y_n = x_n + c$,what are the values of sample mean and median of y_i?

Exercises

- If we add a constant c to each of sample data, $y_1 = x_1 + c, y_2 = x_2 + c, y_3 = x_3 + c, \ldots y_n = x_n + c$,what are the values of sample mean and median of y_i?

$$\bar{y} = \frac{\Sigma y_i}{n} = \frac{\Sigma(x_i + c)}{n} = \frac{\Sigma x_i}{n} + \frac{nc}{n} = \bar{x} + c$$

$$\tilde{y} = \text{the median of } (x_1 + c, x_2 + c, \ldots, x_n + c) = \text{median of}$$

$$(x_1, x_2, \ldots, x_n) + c = \tilde{x} + c$$

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Exercises

- If we multiply each x_i by the constant c, $y_i = x_i * c$,what are the value of mean and median of y_i?

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Exercises

- If we multiply each x_i by the constant c, $y_i = x_i * c$,what are the value of mean and median of y_i?

$$\bar{y} = \frac{\Sigma y_i}{n} = \frac{\Sigma(x_i \cdot c)}{n} = \frac{c\Sigma x_i}{n} = c\bar{x}$$

$$\tilde{y} = (cx_1, cx_2, \ldots, cx_n) = c \cdot median(x_1, x_2, \ldots, x_n) = c\tilde{x}$$

Home work

- Textbook exercise of Chapter 1: 14, 20, 33, 44, 55